Oil Prices and Currency Volatility: Time-Varying Evidence from Eight Advanced Economies

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Abstract

How stable is the relationship between oil prices and currency volatility? We study eight advanced economies over March 2000–December 2024 using monthly volatility constructed from daily exchange rates. A local-linear fixed-effects model allows associations with real oil price changes, oil variability, the VIX and US yield changes to evolve over time. Dependence-aware simultaneous bands distinguish changing point estimates from supported sign changes. The common oil coefficient crosses zero, but its baseline band includes zero throughout the interior and accommodates a constant coefficient. Alternative benchmarks and conditioning sets reveal some negative episodes without establishing a positive-to-negative reversal. The VIX association is more clearly positive and time-varying, and country oil slopes depart from their average. Generalized responses from a monthly global-variable system indicate short-horizon financial-risk comovement; oil responses are less precise and sensitive to contemporaneous allocation. Matched-sample local projections provide no stronger oil-sign evidence. The results favour a conditional, currency-specific interpretation of oil–FX linkages, with a clear distinction between contemporaneous associations and joint market dynamics.

Keywords: Oil prices; Exchange-rate volatility; Time-varying coefficients; Simultaneous inference; Generalized impulse responses.
JEL Codes: C14; C22; C23; F31; Q43.

1 Introduction

An increase in the price of oil need not carry the same information for currency markets in every episode. It can accompany stronger world demand, a disruption in supply, or a change in investors' assessment of risk. These circumstances can have different implications for uncertainty about exchange rates, even when the observed oil price moves in the same direction. The empirical question is therefore not simply whether oil and exchange rates are related, but whether their relationship with exchange-rate volatility is sufficiently stable and precisely estimated to support a general economic interpretation.

This distinction matters for both the object of study and the strength of the conclusions. An exchange-rate appreciation is a directional movement in a price; volatility measures the magnitude of fluctuations without identifying their direction. Evidence that oil predicts a currency's level or return is consequently not evidence that it raises or lowers currency risk. Likewise, a coefficient curve crossing zero is not by itself evidence of a statistically significant reversal. Flexible estimators can reveal movements hidden by

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a constant coefficient, but their uncertainty must be assessed over the same time horizon used to tell the economic story.

The literature provides strong reasons to allow for changing relationships, while also cautioning against a simple causal reading. Beckmann et al. (2020) review theoretical channels operating in both directions and evidence that depends on the country, horizon and empirical design. Chen et al. (2010) investigate whether exchange rates predict commodity prices, whereas Ferraro et al. (2015) examine oil-based exchange-rate forecasting. These studies concern prices and predictability; the present analysis instead studies the contemporaneous association with measured FX volatility. More fundamentally, Kilian (2009) shows that oil price movements can reflect distinct underlying shocks. Changes in reduced-form coefficients may therefore reflect changing shock composition, changing transmission, omitted common influences, or sampling variation. A time-varying regression alone cannot distinguish these explanations.

A complementary literature studies the dependence structure of oil and currency returns. Aloui et al. (2013) use a copula–GARCH framework to examine conditional and extreme dependence, while Reboredo (2012) documents generally weak comovement that increases after the global financial crisis. These designs address joint return distributions and tail behaviour. Our object is the conditional mean of measured currency volatility and how its oil-price association evolves over calendar time. This distinction connects the analysis to risk management without treating return correlation, volatility sensitivity and structural transmission as the same empirical quantity.

We use monthly observations for Australia, Canada, Japan, New Zealand, Norway, Sweden, Switzerland and the United Kingdom over March 2000–December 2024. Volatility is constructed from daily bilateral exchange rates against the US dollar. The baseline relates log volatility to the monthly change in the real WTI oil price, a rolling oil volatility proxy, log VIX and the change in the US ten-year government yield. Persistence diagnostics motivate using changes in the oil price and yield rather than treating their levels as ordinary stationary regressors. The resulting oil coefficient measures the association with a monthly price movement; it is not a long-run elasticity with respect to the oil price level.

Our empirical strategy builds on the local-linear varying-coefficient panel framework of Li et al. (2011). Related energy applications include Hailemariam et al. (2019), who study oil prices and economic policy uncertainty. In our setting, all baseline regressors are common across countries. This feature makes the time dimension central to identification: repeating a global oil price eight times in a month does not create eight independent oil price movements. We exploit the exact equivalence between pooled local estimation with country dummies and local estimation of the cross-country mean, and conduct inference over months while retaining cross-country dependence. Smooth coefficient paths are estimated jointly with a smooth intercept, using bandwidths chosen by blocked time validation.

Our analysis contributes in three ways. First, it links a daily-based measure of monthly currency volatility to a flexible oil-price relationship, and quantifies how changing the volatility definition alters the estimated path. Second, it separates pointwise patterns from whole-path sign evidence and coefficient constancy while recognising that common global regressors supply time-series, rather than independent country-level, variation. Third, it connects country heterogeneity with a complementary analysis of global market dynamics. Generalized responses and matched-sample local projections distinguish joint adjustment from the contemporaneous association that holds financial conditions fixed. Together, these exercises clarify which features of the oil–currency relationship are precisely supported and which depend on measurement or conditioning choices.

The evidence places limits on a general narrative of oil-driven volatility reversals. The full-sample constant oil-change coefficient is -0.203 with a HAC standard error of 0.150. Its time-varying counterpart ranges from -0.776 to 0.683 over the interior sample, yet the simultaneous band includes zero

throughout and accommodates a constant coefficient. The VIX coefficient is positive at 92 interior months under simultaneous inference, and its constancy is rejected. Country oil slopes also depart from the cross-country average. In the dynamic system, positive VIX innovations accompany a more consistently positive short-horizon volatility response than oil innovations. The post-GFC oil response has one significantly negative horizon under joint inference, while the full-sample response remains imprecise. These findings motivate a conditional, currency-specific account of oil–FX comovement rather than a universal causal sign reversal.

Section 2 develops the empirical framework and inference. Section 3 describes the data, diagnostics and constant-coefficient benchmarks. Section 4 examines time-varying paths, country heterogeneity, specification sensitivity and dynamic adjustment. Section 5 concludes. Appendices A–D document data construction, local inference, sensitivity design and dynamic methods, respectively.

2 Econometric Framework and Inference

2.1 Estimand, identification and the role of country effects

Let $i = 1, \dots, N$ index countries and $t = 1, \dots, T$ index months, with $N = 8$, $T = 298$, and rescaled time $\tau_t = t/T$. Our baseline is

$$y_{it} = \alpha_i + a(\tau_t) + \beta_g(\tau_t) g_t + \beta_v(\tau_t) v_t + \beta_z(\tau_t) z_t + \beta_q(\tau_t) q_t + \varepsilon_{it}, \qquad \sum_{i=1}^{N} \alpha_i = 0. \tag{1}$$

Here y_{it} is log annualised daily-based FX volatility, g_t is the change in log real WTI, v_t is the oil volatility proxy, z_t is log VIX, and q_t is the change in the US ten-year yield in percentage points. The country effect α_i absorbs persistent differences in average log volatility. The smooth intercept $a(\tau)$ absorbs low-frequency movements not attributed to the included covariates. Each $\beta_j(\tau)$ describes a local contemporaneous association conditional on the other regressors. The linear dependence on covariates remains a restriction; the nonparametric component is their smooth evolution over calendar time.

This specification is useful when relationships evolve gradually but the dates and number of changes are unknown. It avoids imposing a single break date or a small set of regimes. It does not, however, estimate abrupt breaks without smoothing them, establish a nonlinear response to the size of an oil movement, or describe the full conditional distribution of volatility. We use it as a flexible conditional-mean approximation and compare its evidence with a constant-coefficient benchmark.

The source of identifying variation deserves particular attention. Write $x_t = (g_t, v_t, z_t, q_t)'$ and $b(\tau) = (a(\tau), \beta_g(\tau), \beta_v(\tau), \beta_z(\tau), \beta_q(\tau))'$. Averaging equation (1) over countries gives

$$\bar{y}_t = (1, x_t') b(\tau_t) + \bar{\varepsilon}_t, \qquad \bar{y}_t = N^{-1} \sum_i y_{it}. \tag{2}$$

For a balanced panel with common regressors and common kernel weights, pooled local least squares with country dummies yields exactly the same common slopes and sum-normalised intercept as local least squares on $\bar{y}_t$. The within-month deviations contribute to the objective but contain no additional variation in x_t. Cross-sectional averaging may reduce idiosyncratic noise; the relevant temporal information remains T months, with precision depending on cross-country covariance. We therefore do not use NT independent observations to justify the uncertainty bands.

Unrestricted month fixed effects cannot be added to this baseline: they would absorb every global regressor. A smooth intercept is a weaker restriction and is not a substitute for controlling all common

shocks. Local identification requires sufficient variation in the regressors within each time window, after allowing for the local time trend. We check the rank and conditioning of those local designs. If countries have different slopes, equation (2) represents an average relationship; Section 4.3 assesses that restriction directly.

2.2 Local estimation and bandwidth choice

At a target time τ, approximate $b(\tau_s)$ by $b_0 + b_1(\tau_s - \tau)/h$. The local estimator solves

$$(\widehat{b}_h(\tau), \widehat{d}_h(\tau)) = \arg\min_{b_0, b_1} \sum_{s=1}^{T} K\left(\frac{\tau_s - \tau}{h}\right) \left[\bar{y}_s - (1, x_s') \left\{b_0 + b_1 \frac{\tau_s - \tau}{h}\right\}\right]^2, \tag{3}$$

where $K(u) = 0.75(1 - u^2)\mathbf{1}\{|u| < 1\}$ is the Epanechnikov kernel and h is the bandwidth in units of the sample span. The vector b_1 is a scaled local derivative, not an additional economic parameter. Regressors are standardised internally to improve numerical conditioning and all reported coefficients are transformed back to their original units. Rank-deficient designs cause the computation to stop; no undisclosed ridge penalty is used.

Bandwidth choice balances temporal resolution against estimation variance. We select h from $\{0.08, 0.10, 0.12, 0.15, 0.18, 0.22, 0.26, 0.30, 0.35, 0.40\}$ using consecutive six-month validation blocks. For each block, its observations for all countries and the three months on either side are excluded from estimation. Country levels are estimated using training months only. The criterion is the mean squared prediction error across held-out country-months. This embargo reduces leakage through nearby observations but is not an assumption that dependence ends after three months. Because training observations can lie on either side of a holdout block, this is retrospective smoothing validation, not a real-time forecasting exercise. Leaving out a country alone would retain the same global regressor sequence and would not provide an adequate time-bandwidth criterion.

The selected bandwidth is $h = 0.12$. Away from the endpoints, its kernel uses approximately $2hT \simeq 72$ months of calendar support. To reduce leading smoothing bias, the reported estimator is

$$\widetilde{b}(\tau) = 2\widehat{b}_{h/\sqrt{2}}(\tau) - \widehat{b}_h(\tau). \tag{4}$$

Under the usual interior smoothness expansion, this combination cancels the leading h^2 bias. It increases variance, which is incorporated by applying the same combined linear operator in every multiplier draw. We restrict formal inference to $\mathcal{I} = \{t : h \le \tau_t \le 1 - h\}$, corresponding to February 2003–December 2021. Endpoint estimates remain visible as descriptive dashed paths; we do not use them to claim a significant post-2021 reversal.

2.3 Dependence-aware uncertainty and coefficient constancy

An interval valid at a prespecified month is not a confidence band for an entire path. The distinction is central when inspecting many months for sign changes, as emphasised in the time-varying coefficient inference literature (Zhou and Wu 2010). Moreover, independent multipliers across months would discard the serial dependence visible in volatility data. We instead use a dependent wild-bootstrap construction motivated by Shao (2010), with a common multiplier at each date for every country.

Let $H_j(t, s)$ denote the combined linear operator in equation (4), so $\widetilde{b}_j(\tau_t) = \sum_s H_j(t, s)\bar{y}_s$. We centre the fitted residuals and adjust their scale using the residual row norm of this linear smoother, as defined in Appendix 5. Denote the adjusted residual by $\widehat{u}_s$. This scaling is a homoskedastic reference

correction, while temporal dependence enters through the multiplier construction. For replication r, generate independent standard normal innovations and construct moving-average multipliers $\xi_s^{(r)}$ with

$$E^*\xi_s^{(r)} = 0, \qquad \mathrm{Cov}^*(\xi_s^{(r)}, \xi_{s-k}^{(r)}) = \left(1 - \frac{|k|}{L}\right)_+, \qquad L = 12. \tag{5}$$

Each multiplier is the sum of L consecutive innovations divided by $\sqrt{L}$. The bootstrap coefficient perturbation is

$$\Delta_j^{(r)}(t) = \sum_s H_j(t,s)\widehat{u}_s\xi_s^{(r)}. \tag{6}$$

The same multiplier applies to every country in a month, preserving cross-country dependence. We use 1,999 replications with a fixed random seed and examine multiplier spans of 6 and 18 months alongside the 12-month baseline. The regressor design and country composition remain fixed during inference.

Let $\widehat{s}_j(t)$ be the bootstrap standard deviation of $\Delta_j^{(r)}(t)$. Pointwise intervals use the empirical 95th percentile of $|\Delta_j^{(r)}(t)/\widehat{s}_j(t)|$ at each date. The simultaneous critical value c_j is the empirical 95th percentile of

$$\max_{t\in\mathcal{I}} \frac{|\Delta_j^{(r)}(t)|}{\widehat{s}_j(t)}. \tag{7}$$

The corresponding band is $[L_j(t), U_j(t)] = \widetilde{b}_j(\tau_t) \pm c_j\widehat{s}_j(t)$. It is simultaneous over interior dates for a given curve, not jointly over all four economic regressors. Inference is conditional on the selected bandwidth and maintained smoothness, local identification and weak-dependence conditions. It does not incorporate bandwidth-selection uncertainty or establish exogeneity; the nominal coverage is an asymptotic approximation, not a finite-sample guarantee.

We also ask a specific inferential question: can any constant coefficient fit the whole simultaneous band? Define

$$\mathcal{C}_j = \bigcap_{t\in\mathcal{I}} [L_j(t), U_j(t)] = \left[\max_{t\in\mathcal{I}} L_j(t),\ \min_{t\in\mathcal{I}} U_j(t)\right]. \tag{8}$$

An empty intersection rejects constancy. This is a conservative test obtained by inverting the band: if a constant true path is covered everywhere, its value necessarily belongs to the intersection. A nonempty intersection therefore means that constancy cannot be rejected by this procedure; it does not establish that the coefficient is constant. Similarly, a claim of a statistically supported sign reversal requires positive and negative intervals under simultaneous inference, not merely a zero crossing of the point estimate.

2.4 Benchmarks and the limits of causal interpretation

A constant-coefficient country fixed-effects regression provides a readily interpretable average association and a check on the magnitudes from the flexible model. Its common slopes equal those from a time-series regression of $\bar{y}_t$ on a constant and x_t. We use Newey and West (1987) covariance estimates with 12 monthly lags for this regression. Working with the country mean allows its time-series covariance to reflect contemporaneous cross-country dependence; the lag correction accommodates serial dependence. Separate pre-GFC, post-GFC and COVID-onward estimates describe how averages differ across broad periods, but are not formal break tests.

Oil prices, financial risk and exchange rates can respond jointly to news. Country effects and robust uncertainty calculations do not distinguish those sources. Conditioning on VIX asks how volatility covaries with oil holding the observed financial-risk proxy fixed, and may absorb part of an underlying transmission channel. Section 4.4 therefore compares information sets and studies joint dynamic innova-

tions separately. A predictive lag of oil is not sufficient to justify an exclusion restriction; no internally instrumented structural oil effect is assumed.

3 Data, Diagnostics and Benchmark Evidence

3.1 Daily-based volatility and global covariates

The balanced panel covers 298 months and eight economies, giving 2,384 country-month observations. Daily bilateral exchange rates come from the Federal Reserve's H.10 series through FRED (Board of Governors of the Federal Reserve System 2026a); Appendix 5 lists every series. Let r_{id} be the change in the log exchange rate between consecutive available daily observations. If D_{it} is the set of return observations ending in month t, and $n_{it} = |D_{it}|$, define

$$RV_{it} = \left(\frac{252}{n_{it}} \sum_{d \in D_{it}} r_{id}^2 \right)^{1/2}, \qquad y_{it} = \log RV_{it}. \tag{9}$$

The series contains 18–23 returns per country-month. Missing holiday quotes are removed before differencing; they are not replaced by zero returns. An interval spanning a holiday is assigned to the month of its ending quote. The factor 252 is an annualisation convention, including for such intervals. Reciprocal quotation conventions reverse return signs but leave their squares unchanged. This is a daily-sampled, annualised volatility measure rather than an intraday integrated-variance estimator. The realised-volatility literature provides the measurement motivation (Andersen et al. 2003); daily sampling does not inherit all the precision available from high-frequency observations.

Two alternative outcomes clarify what daily sampling contributes. One is the demeaned standard deviation of within-month daily returns. The other is the annualised three-month sample standard deviation of month-end FX log returns, constructed from the same daily quotes. The latter uses only three monthly observations and overlapping windows. Its pooled mean is 9.06% annualised, compared with 10.01% for the daily-based measure, and their pooled log correlation is 0.404. These measures capture different aspects of currency variability; Appendix 5 provides their exact definitions.

Nominal WTI is the IMF monthly period-average dollar price, distributed as FRED series `POILWTIUSDM` (International Monetary Fund 2026b). We deflate it by the US all-items seasonally adjusted CPI, `CPIAUCSL` (U.S. Bureau of Labor Statistics 2026), and normalise real dollar values to January 2000. Thus

$$p_t^R = \log\left(P_t^{WTI} \frac{CPI_{2000:01}}{CPI_t} \right), \qquad g_t = \Delta p_t^R, \qquad v_t = \sqrt{12}\, \mathrm{sd}_{j=0,1,2}(\Delta \log P_{t-j}^{WTI}). \tag{10}$$

The sample standard deviation in v_t uses all three monthly observations and measures rolling monthly oil variability, not daily realised oil volatility. Its overlapping construction is one additional reason to allow serial dependence. The positive monthly average oil-price series makes the logarithm well-defined; no logarithm of a negative daily futures settlement is used.

The VIX is the log of the arithmetic monthly mean of available daily closing observations in `VIXCLS` (Cboe 2026). It represents US equity-market implied volatility and associated risk conditions. The yield variable is $q_t = \Delta R_t^{10}$, where R_t^{10} is the monthly average US ten-year Treasury yield in percent, series `GS10` (Board of Governors of the Federal Reserve System 2026c). Additional specifications use IMF Brent prices and Federal Reserve US industrial-production growth (International Monetary Fund 2026a, Board of Governors of the Federal Reserve System 2026b).[1]

[1] All raw downloads are frozen at 9 November 2026, and the estimation sample ends in December 2024.

Table 1: Summary statistics, March 2000–December 2024

Variable	Obs.	Mean	SD	Minimum	Maximum
Annualised FX volatility (%)	2,384	10.008	4.534	2.760	68.724
$y_{it} = \log RV_{it}$	2,384	-2.380	0.385	-3.590	-0.375
Real oil price change, g_t	298	0.001	0.100	-0.541	0.529
Oil volatility proxy, v_t	298	0.250	0.203	0.008	2.148
Log VIX, z_t	298	2.922	0.348	2.315	4.138
US yield change (pp), q_t	298	-0.007	0.218	-1.110	0.650

Notes: FX volatility is reconstructed from daily exchange-rate observations and is annualised. The panel contains eight countries and 298 months. Global regressors are summarised over 298 distinct months, not over their eight repeated country copies. SD denotes the sample standard deviation; pp denotes percentage points. Exact definitions and sources appear in Section 3.

Table 1 reports variable magnitudes using 298 distinct observations for each global series. Monthly oil changes average close to zero but have a standard deviation of about 0.10 log points, making even moderate coefficients potentially economically relevant. Yield changes are measured in percentage points, so one unit represents 100 basis points. The largest measured currency volatility is 68.72% annualised for Australia in October 2008. The sample retains such stress observations, which are central to the variation in currency risk studied here.

Figure 1 places the four markets in context. Currency volatility and the VIX exhibit pronounced stress episodes, while oil prices and bond yields display persistent movements over longer horizons. The gap between the daily and monthly-return FX measures changes over time, so replacing one with the other is more than a rescaling. The combination of common stress episodes and differing low-frequency behaviour motivates controlling for the financial environment, checking persistence and allowing conditional relationships to evolve.

3.2 Persistence and dependence diagnostics

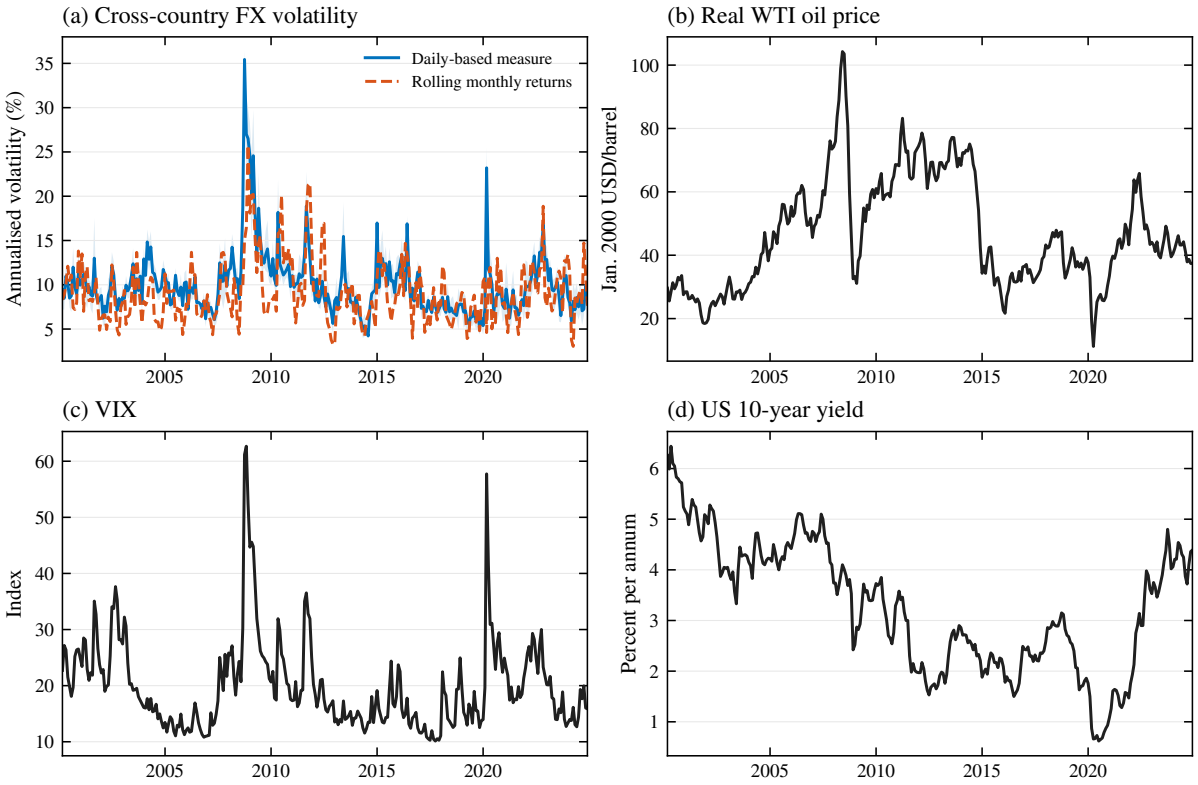
Table 2: Persistence and residual dependence

	ADF			KPSS	
Series	Statistic	p-value	Lags	Statistic	p-value
Mean log FX volatility	-5.123	< 0.001	1	0.128	0.084
Log real WTI price	-2.737	0.221	2	0.398	≤ 0.010
Real oil price change	-10.447	< 0.001	2	0.057	≥ 0.100
Oil volatility proxy	-5.845	< 0.001	4	0.062	≥ 0.100
Log VIX	-4.533	0.001	0	0.107	≥ 0.100
US 10-year yield	-1.852	0.679	2	0.354	≤ 0.010
US yield change	-11.944	< 0.001	1	0.297	≥ 0.100
Dependence diagnostics (model residuals)					
Mean pairwise correlation, raw y					0.638
Mean pairwise residual correlation					0.231
Country residual AR(1), range					0.123–0.502

Notes: The ADF null is a unit root; the KPSS null is stationarity around the specified deterministic terms. Level-series tests include an intercept and trend; change-series tests include an intercept. ADF lag order is selected by AIC from 0–12; KPSS uses an automatic Bartlett bandwidth. KPSS inequalities reflect tabulated probability limits. The level-series sample contains 298 months; change-series tests difference those observations (297 months). Dependence statistics are descriptive and use the full-sample fitted residuals; they do not certify exogeneity or correct specification.

Table 2 combines augmented Dickey–Fuller tests with the complementary stationarity-null test of Kwiatkowski et al. (1992). The level-series tests allow for a deterministic trend; change-series tests

Figure 1: Currency volatility and global market conditions



Notes: Panel (a) reports the cross-country arithmetic mean of annualised FX volatility; the shaded area is its cross-country interquartile range, not a confidence interval. The dashed line is the three-month standard deviation of month-end FX returns. Other panels show levels for economic context; the regressions use changes in log real WTI and in the yield. All panels use separate, explicitly labelled scales.

include a constant. The ADF test does not reject a unit root in log real oil or the ten-year yield ($p = 0.221$ and 0.679), while KPSS rejects trend stationarity at its 1% tabulation limit for both. Their first differences give the opposite pattern. These tests do not prove a particular integration order, especially in the presence of breaks, but they give a concrete reason not to attach ordinary stationary-regression confidence bands to the levels specification. The baseline therefore uses g_t and q_t. The oil volatility proxy and log VIX reject the ADF unit-root null and do not reject the corresponding KPSS null at conventional levels. Mean log FX volatility also rejects the ADF null, although its KPSS p-value of 0.084 gives less decisive support for a stable mean. The maintained approximation is locally stable dependence around smooth coefficient functions. Allowing the intercept and slopes to evolve accommodates gradual changes in conditional relationships; it does not dispense with the dependence and smoothness conditions needed for inference.

Dependence is economically and statistically material. The mean pairwise correlation of log FX volatility is 0.638. After fitting the common time-varying model, the average residual correlation remains 0.231 and country residual first-order autocorrelations range from 0.123 to 0.502. Thus, neither the cross-section nor successive months can be treated as independent replications. These diagnostics motivate common temporal multipliers and HAC benchmark errors. Residual dependence also leaves room for omitted common influences; robust covariance estimation accommodates dependence in uncertainty calculations but does not remove omitted-variable bias.

3.3 What do constant coefficients reveal?

Table 3: Constant-coefficient country fixed-effects benchmarks

	Full sample	Pre-GFC	Post-GFC	COVID onward
Constant	-3.772^{***}	-2.745^{***}	-4.356^{***}	-3.588^{***}
	(0.349)	(0.356)	(0.374)	(0.445)
Real oil price change	-0.203	0.280	0.055	-0.381^{**}
	(0.150)	(0.225)	(0.244)	(0.162)
Oil volatility proxy	0.102	0.074	0.428^{*}	0.046
	(0.123)	(0.173)	(0.224)	(0.101)
Log VIX	0.468^{***}	0.117	0.656^{***}	0.383^{**}
	(0.126)	(0.115)	(0.131)	(0.165)
US yield change (pp)	0.053	-0.020	0.302^{**}	-0.023
	(0.060)	(0.087)	(0.152)	(0.094)
Months	298	102	127	59
Country-month observations	2,384	816	1,016	472
Within R^2	0.233	0.027	0.279	0.210
Country fixed effects	Yes	Yes	Yes	Yes

Notes: Pre-GFC: March 2000–August 2008; post-GFC: July 2009–January 2020; COVID onward: February 2020–December 2024. September 2008–June 2009 is included only in the full-sample column. Standard errors in parentheses use Newey–West HAC covariance on the cross-country-mean regression, with 12 lags, Bartlett weights and the $T/(T-5)$ finite-sample factor. This exploits the exact common-regressor equivalence and allows contemporaneous dependence across countries and serial dependence over months. The constant uses a sum-to-zero fixed-effect normalisation. Within R^2 is computed from the country panel after removing country means. Stars are normal-approximation significance at 10% (*), 5% (**) and 1% (***). Regime contrasts are descriptive, not structural-break tests.

Table 3 provides a scale benchmark before imposing temporal flexibility. In the full sample, the oil-change estimate is -0.203 (standard error 0.150), so the data do not reject a zero average coefficient at the 5% level. Relative to an unchanged real oil price, a 10% price increase gives $g_t = \log(1.10)$ and an associated volatility difference of $100[\exp\{-0.203\log(1.10)\} - 1] \simeq -1.91\%$, holding the other regressors fixed. This is a proportional difference in the volatility measure, not a 1.91-percentage-point reduction

in the exchange rate or its annualised volatility.

The VIX coefficient is 0.468 (standard error 0.126). A 10% higher monthly VIX is associated with approximately 4.56% higher FX volatility under the same calculation. This is both more precisely estimated and more consistently related to the volatility outcome than the average oil-change coefficient. The full-sample oil-volatility and yield-change coefficients are imprecise. The within-panel R^2 is 0.233; it describes in-sample fit after country demeaning, not the fraction of volatility causally generated by the included markets.

The broad-period estimates suggest that a single average can conceal important differences, but also illustrate why regime comparisons need care. The oil-change coefficient is 0.280 before the GFC, 0.055 in July 2009–January 2020, and −0.381 from February 2020 onward. Only the last is individually significant at 5% using the reported HAC approximation. The VIX coefficient rises from an imprecise 0.117 before the GFC to 0.656 in the post-GFC period. A significant estimate in one subsample and an insignificant estimate in another is not itself a significant difference between them. The regimes also exclude September 2008–June 2009 from the two neighbouring subsamples, while the full sample retains those months. We consequently use these columns as motivation for continuous time variation, and rely on simultaneous-band inversion for the formal constancy question.

4 Time-varying Evidence

4.1 Smoothing choices and inferential scope

The empirical analysis now asks whether the apparent variation in average associations survives uncertainty about a whole coefficient path. Blocked validation chooses $h = 0.12$; Appendix Figure ?? reports the complete criterion over the prespecified grid. The selected model uses a smooth intercept and all four time-varying regressors jointly. The median and maximum condition numbers of the standardised weighted local design are 8.27 and 11.33 on the inference region; the narrower fit used for bias correction has a maximum of 15.67. Every design has full rank. These calculations rule out a numerical singularity in the reported estimates, but do not establish a causal identifying restriction.

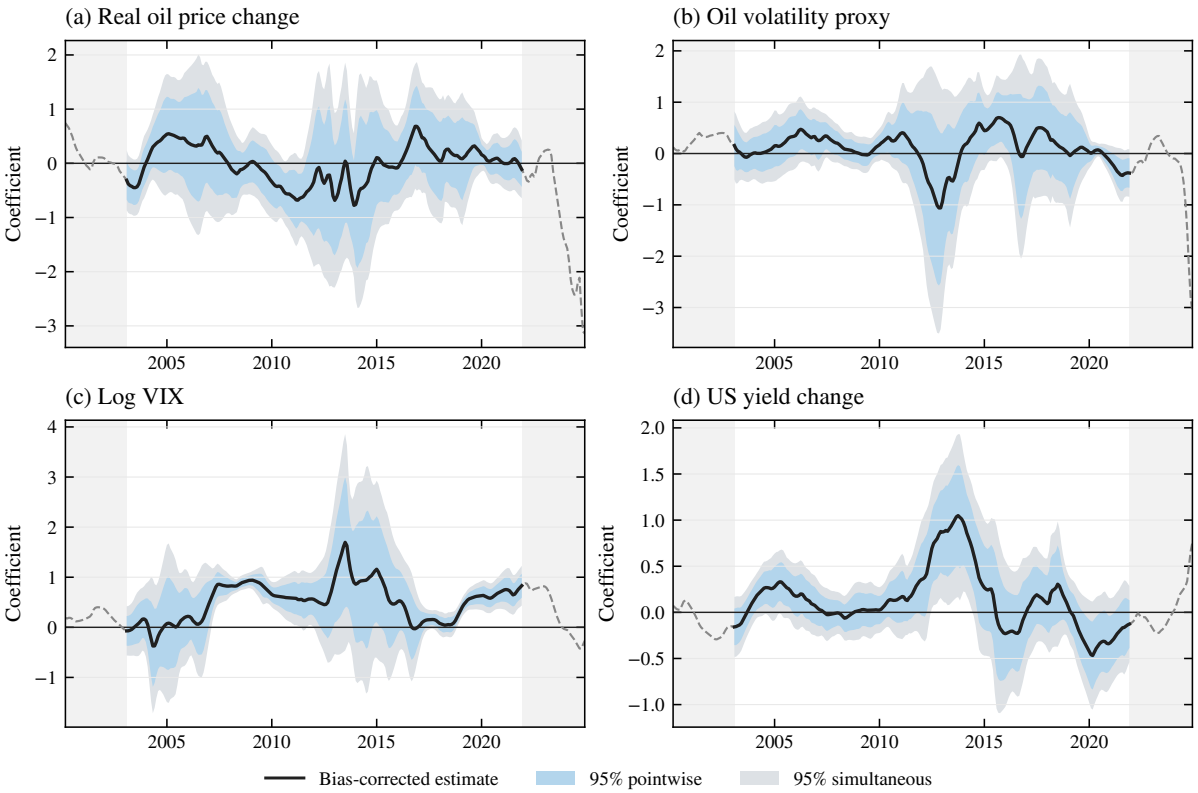
The reported confidence levels are 95% throughout. Figure 2 distinguishes the narrower pointwise intervals from the wider bands simultaneous over February 2003–December 2021. Its grey endpoint regions are displayed for continuity but excluded from formal sign and constancy claims. The interior restriction is substantively consequential: the data through 2024 inform nearby fits, yet the present bandwidth does not support an interior-band claim about the 2022–2024 inflation episode. Likewise, the estimated paths are retrospective and cannot be read as forecasts that would have been available at each historical date.

4.2 Coefficient paths: economic magnitudes versus sign evidence

The oil-change path in Figure 2(a) is positive over parts of the mid-2000s, negative for much of the early 2010s and again positive around 2016–2017. Its interior minimum is −0.776 in December 2013 and maximum is 0.683 in November 2016. At those point estimates, a log(1.10) increase in the monthly real oil change corresponds to roughly 7.13% lower and 6.72% higher volatility, respectively. Those magnitudes illustrate why the path is economically interesting; the dates are estimated extrema and are not independently selected events or precisely dated structural breaks.

The central inferential finding is more restrained. Table 4 records 25 months with negative pointwise oil intervals and none with positive pointwise intervals. Once uncertainty is assessed simultaneously over

Figure 2: Time-varying conditional associations and uncertainty



Notes: Black: bias-corrected coefficient estimates. Blue shading: 95% pointwise intervals. Wider grey shading: 95% bands simultaneous over interior dates for each coefficient separately. Grey vertical endpoint regions and dashed paths are descriptive only. Bands use 1,999 dependent Gaussian multiplier draws, span 12 and the fixed selected bandwidth $h = 0.12$. Coefficient units differ across panels; the oil regressor is a monthly log change, not a log price level.

Table 4: Time variation, sign evidence and coefficient constancy

	Coefficient path			Pointwise		Simultaneous		Compatible
Variable	Min.	Max.	Mean	+	−	+	−	constant set
Oil change	-0.776	0.683	-0.008	0	25	0	0	$[-0.135, 0.037]$
Oil volatility	-1.060	0.700	0.104	33	10	0	0	$[-0.053, 0.062]$
Log VIX	-0.373	1.697	0.515	125	0	92	0	$\varnothing$
Yield change	-0.468	1.049	0.135	51	15	24	0	$\varnothing$

Notes: All entries use the 227 interior months February 2003–December 2021 and the bias-corrected estimator. + and − count months whose 95% interval excludes zero above or below it. Simultaneous bands control over time separately for each curve, not jointly across the four regressors. The compatible constant set is the intersection of simultaneous intervals over all interior months; an empty set rejects constancy at the nominal 5% level under the maintained band approximation. A nonempty set does not establish constancy. The mean is an unweighted time average, not the constant-coefficient regression estimate. Inference uses 1,999 dependent Gaussian multiplier draws and span 12.

the interior, no month excludes zero in either direction. Moreover, constants between approximately -0.135 and 0.037 fit the entire simultaneous band. The data therefore do not establish either a sign reversal or rejection of a constant oil-change association under this design. This is not evidence that oil is irrelevant: the bands allow economically nontrivial values and the average relationship may mask country differences. It is evidence against treating a visually changing curve as a precisely estimated universal transmission channel.

The oil-volatility path in panel (b) similarly contains substantial point-estimate variation. It ranges from -1.060 in November 2012 to 0.700 in September 2015. Because v_t is an annualised decimal proxy rather than a logarithm, a 0.10 increase in it corresponds at the positive maximum to approximately $100[\exp(0.700 \times 0.10) - 1] = 7.25\%$ higher FX volatility. However, 33 positive and ten negative pointwise months become zero significant months under simultaneous inference. Its compatible constant set is also nonempty. The correct reading is an imprecisely estimated changing association with rolling oil variability, not evidence that oil uncertainty reliably switches from stabilising to destabilising currencies.

The VIX results are stronger. Panel (c) shows a positive relationship across substantial portions of the post-GFC sample, with a maximum point estimate of 1.697 in July 2013. A 10% increase in VIX at that point estimate is associated with about 17.6% higher measured FX volatility, although the local interval is wide and the peak should not be treated as a precisely calibrated policy parameter. There are 125 positive pointwise months and 92 positive simultaneous-band months, with no negative significant months. The constant-set endpoints are 0.782 and 0.274 in reversed order, so the intersection is empty and constancy is rejected. Thus the evidence supports temporal variation in the strength of the risk-proxy association more clearly than a change in its sign. This association is consistent with currency-market risk responding to a broader financial environment; it does not identify the VIX as an exogenous shock.

The yield-change coefficient is also time-varying according to the band-inversion test. Its simultaneous band is positive for 24 months, July 2012–June 2014, but never significantly negative. The positive peak is 1.049 in October 2013. Since the yield change is measured in percentage points, a 25-basis-point increase corresponds at this peak to $100[\exp(1.049 \times 0.25) - 1] \simeq 30.0\%$ higher volatility. This large local point estimate should be read with its uncertainty and the limited support of monthly yield movements in mind. The regression cannot determine whether it reflects monetary-policy news, changing risk premia, common news or omitted influences. A negative estimate elsewhere is not evidence of a reliably stabilising yield effect.

4.3 Common movements and cross-country heterogeneity

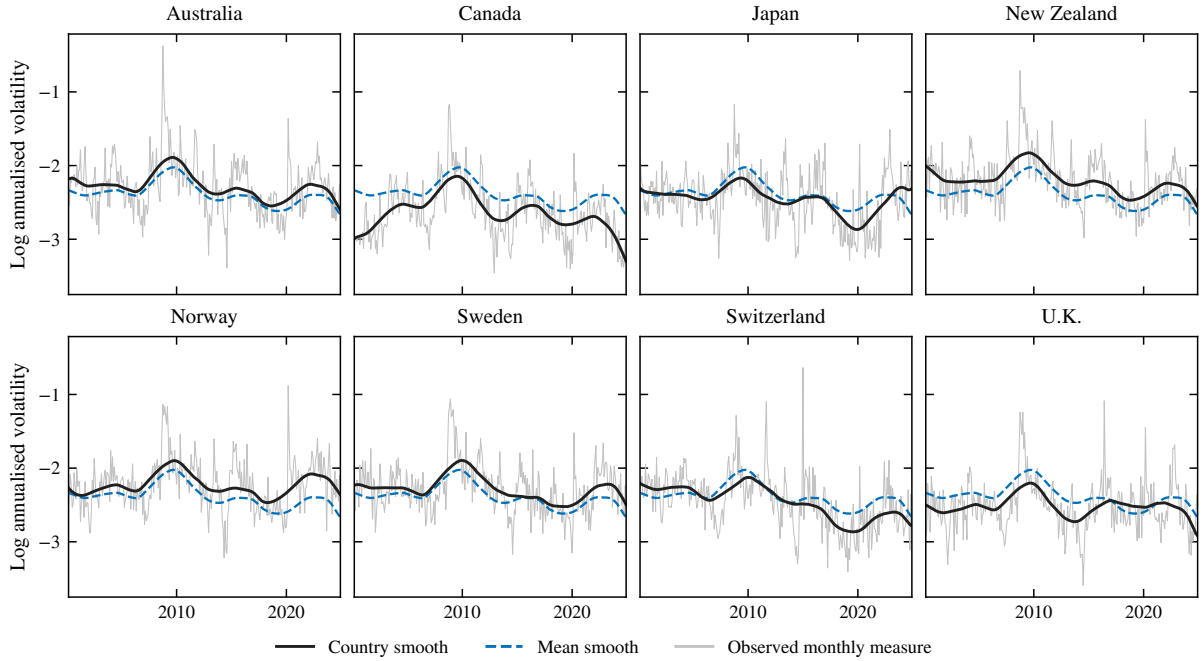
Countries can share broad volatility episodes without having identical conditional oil sensitivities. We therefore separate a descriptive trend comparison from a slope-heterogeneity exercise. For the former, smooth each country's y_{it} against time alone using a local-linear bandwidth of 0.10. Let $s_i(\tau_t)$ denote that smooth and $c(\tau_t) = N^{-1} \sum_i s_i(\tau_t)$ its cross-country mean. Define

$$\lambda_i = T^{-1} \sum_t \{s_i(\tau_t) - c(\tau_t)\}, \qquad m_i(\tau_t) = s_i(\tau_t) - c(\tau_t) - \lambda_i. \tag{11}$$

Then $\sum_i \lambda_i = 0$, $\sum_i m_i(\tau_t) = 0$ at each date, and $\sum_t m_i(\tau_t) = 0$ for each country. This normalisation distinguishes persistent country levels from smooth deviations. The unconditional common smooth $c(\tau)$ is not the conditional intercept $a(\tau)$ in equation (1); the former includes movements associated with all observed and unobserved influences.

Figure 3 shows both shared timing and persistent differences in levels. Mean annualised volatility is

Figure 3: Country volatility paths and their common smooth



Notes: Grey: observed log daily-based FX volatility. Black: country-specific local-linear time smooth, bandwidth 0.10. Dashed blue: the cross-country average of those smooths. All panels use the same scale. These are descriptive, unconditional smooths, with no confidence-band or structural-factor interpretation.

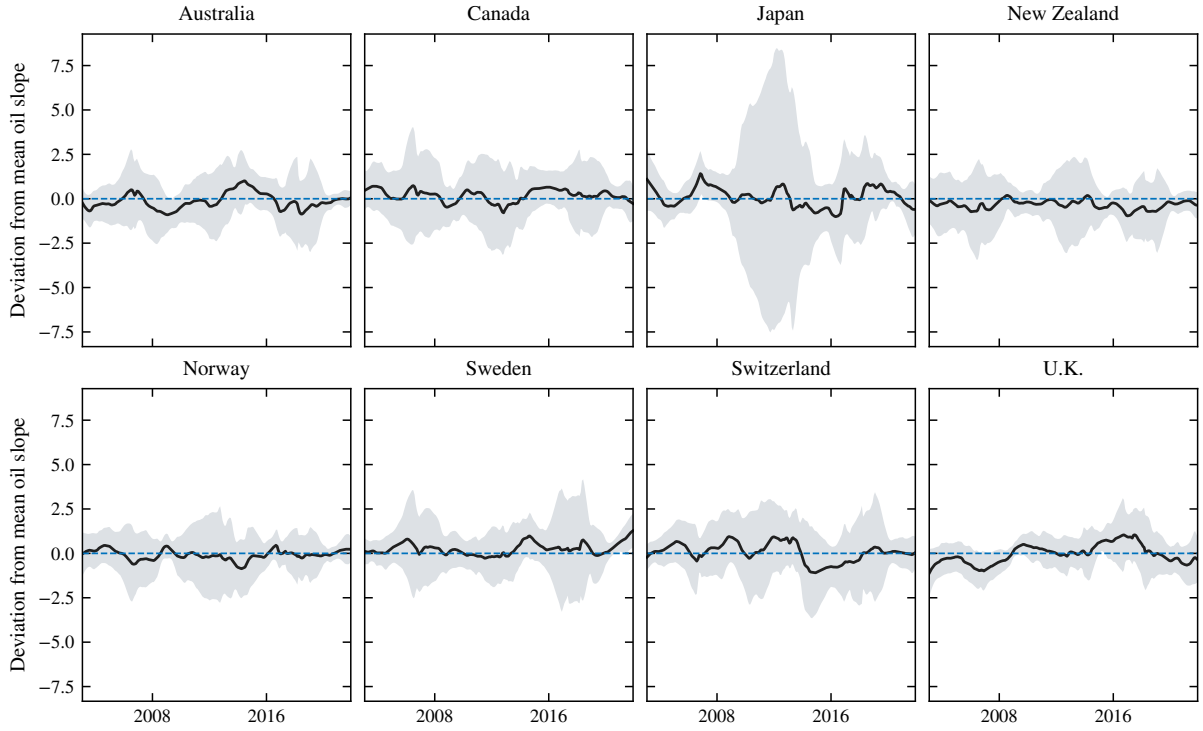
highest for New Zealand (11.75%) and Norway (11.47%) and lowest for Canada (7.80%) and the United Kingdom (8.65%). These are arithmetic means of measured volatility, not fixed effects exponentiated into causal country premiums. The common timing is substantial: unrestricted month indicators account for 67.9% of the sum of squares after country demeaning. This saturated descriptive statistic is not the variance share explained by oil, the VIX, the plotted smooth, or an identified global factor. A common US-dollar denominator is itself a reason to expect comovement.

To evaluate the common-slope restriction more directly, we estimate the same local model separately for each country, allowing all four slopes and the intercept to vary by country. With common design and bandwidth, the average estimated coefficient equals the coefficient estimated from the country mean. Let $d_i(\tau) = \widetilde{\beta}_{g,i}(\tau) - N^{-1}\sum_k \widetilde{\beta}_{g,k}(\tau)$ measure the oil-slope deviation. The multiplier procedure preserves covariance across countries and constructs one maximum-t critical value over both the eight countries and all interior months. The resulting bands therefore address departures from a common oil slope without applying eight unadjusted time-path tests.

Figure 4 rejects the restriction that all countries share the same oil-change slope at every interior date. Under the joint critical value of 3.716, Sweden lies above the mean slope for 16 months and Japan for five; the United Kingdom has four months above and one below, while Australia has one below. Appendix Table ?? reports all countries. These findings do not contradict the absence of a significant average oil coefficient: some currencies may deviate from an imprecise or near-zero average. A positive deviation also need not imply a positive country coefficient.

The heterogeneous paths identify a limit to pooling, while leaving the economic source of those differences open. The data contain no time-varying measure of net oil exports, energy dependence, hedging exposure or intervention that would distinguish competing country mechanisms. The panel also represents a fixed group of eight economies rather than a random sample of currencies. The combination

Figure 4: Country oil-change slopes relative to the cross-country average



Notes: Lines show $d_i(\tau)$, the country oil-change coefficient less the mean coefficient. Shading is a 95% band simultaneous over eight countries and 227 interior dates, constructed from common dependent multipliers. The horizontal reference is equality with the average slope, not a zero country slope. All country regressions allow the other three covariate slopes and intercept to vary.

of common volatility episodes and heterogeneous oil sensitivities therefore motivates separate checks on the average relationship, country composition and joint global dynamics.

4.4 Sensitivity and joint market dynamics

The oil coefficient can change because the economic relationship changes, but also because the oil benchmark, volatility measure or conditioning information changes. Moreover, a contemporaneous coefficient does not describe how jointly evolving markets adjust over subsequent months. This section examines these issues separately. The first set of exercises re-estimates the conditional oil coefficient under alternative measurements and information sets. The second studies innovations in the joint dynamics of global variables and average currency volatility. Keeping these questions distinct is essential: a dynamic innovation response need not equal a coefficient that holds contemporaneous financial conditions fixed.

4.4.1 Measurement, conditioning information and sample composition

Table 5 reports alternatives chosen to address specific features of the baseline. Brent provides a different international oil benchmark; it replaces both the real oil change and the rolling oil-volatility regressor. The demeaned daily standard deviation separates daily sampling from the choice between a second moment and a centred dispersion measure. A three-month standard deviation of month-end FX returns tests a more substantial change in the volatility concept. All three measures are constructed from documented observations rather than treated as interchangeable labels for currency risk.

The demeaned daily measure is close to the baseline: the average interior oil coefficient is 0.006 rather

Table 5: Oil-coefficient sensitivity to measurement, conditioning and sample

		Oil-change coefficient			SCB sign	
Specification	Months	Mean	Minimum	Maximum	+	−
Baseline	298	-0.008	-0.776	0.683	0	0
Brent benchmark	298	-0.237	-2.344	0.698	0	1
Demeaned daily SD	298	0.006	-0.677	0.631	0	0
Without VIX	298	-0.220	-1.911	1.197	0	11
Add US industrial production	298	0.000	-0.789	0.713	0	0
Add lagged mean volatility	297	-0.069	-1.081	0.650	0	0
Through December 2019	238	-0.067	-0.776	0.683	0	0
Rolling monthly returns	298	0.074	-2.379	1.732	0	0

Notes: Each row refits all coefficients and the intercept. SCB sign columns count dates whose 95% time-simultaneous band excludes zero. No row rejects oil-coefficient constancy by band intersection. Inference uses 1,999 dependent multiplier draws with span 12 and the baseline calendar half-window of about 36 months. The main interior is February 2003–December 2021 (227 dates); the lagged-outcome interior starts March 2003 (226), and the pre-2020 sample ends its interior in December 2016 (167). The means and extrema therefore refer to the stated row-specific interiors, not an identical set of dates. Without-VIX and lagged-outcome regressions change the conditioning set. Brent replaces both the oil-price-change and oil-volatility regressors. Comparisons are sensitivity exercises, not a joint test across specifications.

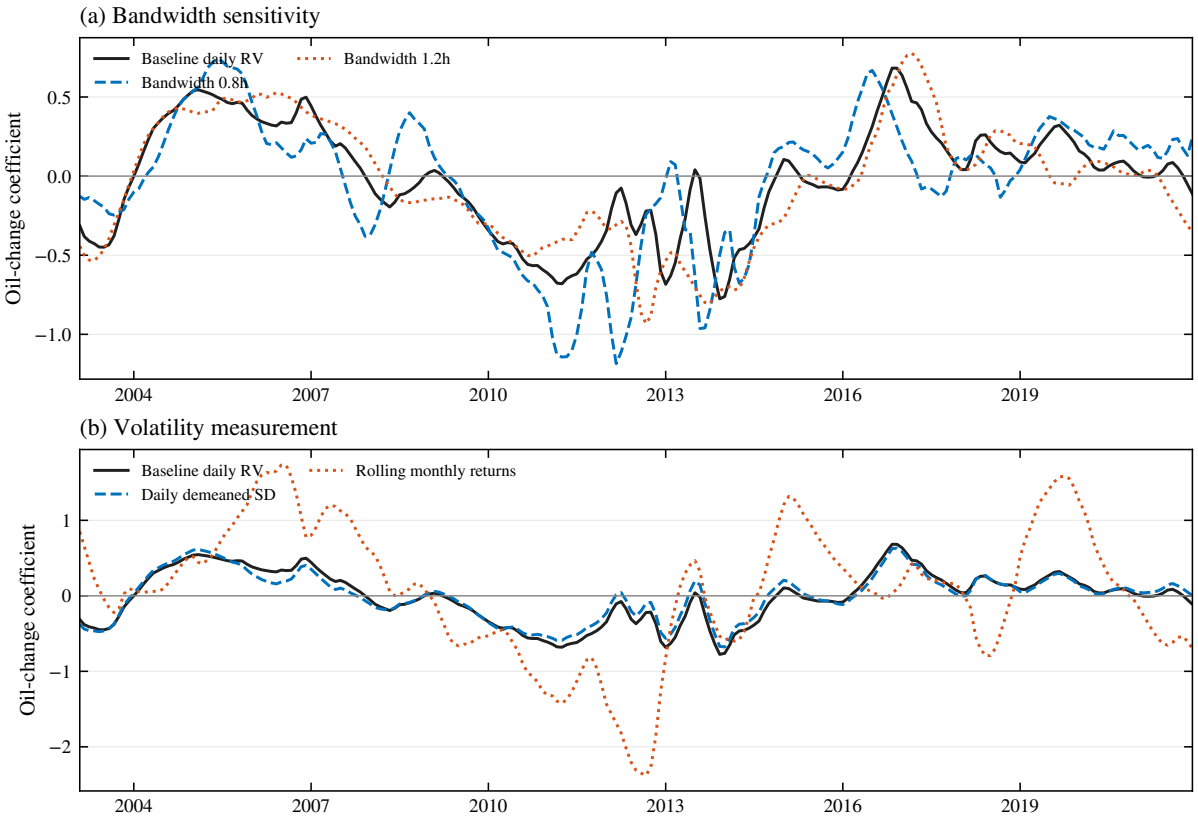
than -0.008, and neither path excludes zero under simultaneous inference. The monthly-return measure produces much larger local amplitudes, ranging from -2.379 to 1.732, but its simultaneous band also includes zero throughout. This difference is economically meaningful. A small number of monthly net returns can conceal large within-month movements and generates overlapping three-month windows; it therefore measures a different component of currency variability. Figure 5(b) shows that measurement changes can alter the apparent depth of an episode even when they do not strengthen the evidence for its sign.

Brent produces a more negative average coefficient, -0.237, and one interior month with a negative simultaneous interval. Nevertheless, a constant coefficient remains compatible with its band, and there is no simultaneously significant positive interval. Thus benchmark choice affects local estimates and isolated sign evidence. The finding common to WTI and Brent is the absence of a statistically established positive-to-negative reversal, rather than equality of their estimated paths or universal absence of negative association.

The next specifications address conditioning information. Removing VIX allows oil movements to reflect the financial-risk comovement that the baseline holds fixed. The mean oil coefficient becomes -0.220, with 11 negative simultaneous-band months. This is a different conditional projection, not an estimate of a causal total effect. Since VIX and oil prices can respond to the same news, neither inclusion nor exclusion by itself resolves endogeneity. The contrast shows why the statement that oil stabilises currencies depends on the information held constant. Adding monthly US industrial-production growth provides a check on observed activity conditions: the mean oil coefficient is approximately zero and the simultaneous band includes zero throughout. US industrial production is a limited activity proxy, not a measure of world oil demand or an exogenous control for every global disturbance.

Adding lagged mean log FX volatility examines whether persistence changes the contemporaneous oil association. Its average oil coefficient is -0.069, with no significant simultaneous interval. This specification conditions on volatility history and therefore estimates a short-run association; it does not furnish an instrument for current oil prices. Estimation through December 2019 also leaves a nonempty compatible-constant set. To preserve the baseline calendar smoothing window, the truncated-sample interior ends in December 2016. This exercise assesses the earlier relationship without pandemic observations and cannot establish a separate coefficient path for the pandemic itself.

Figure 5: Sensitivity of the conditional oil-change path



Notes: Panel (a) changes the baseline bandwidth by ±20%. Panel (b) compares daily second-moment volatility, a demeaned daily standard deviation and the three-month standard deviation of month-end returns. Curves use identical baseline global covariates and are displayed over the baseline interior. They are point estimates; formal sign counts use each specification's own interior and bands in Tables 5 and ??.

Smoothing and country composition provide complementary checks. Figure 5(a) shows that changing the bandwidth affects local amplitudes and the timing of extrema. Neither the smaller nor the larger bandwidth rejects oil-coefficient constancy. Omitting one country at a time produces interior mean coefficients between -0.044 and 0.040. Omitting Canada yields three negative simultaneous-band months; the other omissions yield none, and every omission retains a nonempty compatible-constant set. Likewise, multiplier spans of 6, 12 and 18 months produce no significant baseline oil interval. Appendix 5 reports these results in full. They support caution about a universal oil sign reversal while documenting, rather than suppressing, specification-sensitive negative episodes. The bands control over time within an exercise; they do not make the collection of robustness specifications a single multiple-testing-adjusted experiment.

4.4.2 Dynamic specification and innovation interpretation

Contemporaneous regressions leave open whether an oil-associated currency movement persists. To examine joint adjustment, define the monthly vector

$$Z_t = (g_t, v_t, z_t, q_t, \bar{y}_t)', \qquad Z_t = c + \sum_{\ell=1}^{2} A_\ell Z_{t-\ell} + u_t, \qquad E(u_t u_t') = \Sigma. \tag{12}$$

Here c is a five-element intercept, A_ℓ contains cross-variable lag coefficients, and u_t is a vector of reduced-form innovations. The outcome $\bar{y}_t$ is the average of log currency volatilities, so its exponential is their geometric mean. Because all four market variables are global, the system contains one observation per month. Country replication cannot supply additional histories of an oil or VIX innovation. Country heterogeneity remains the object of Section 4.3; the dynamic system concerns the average currency response.

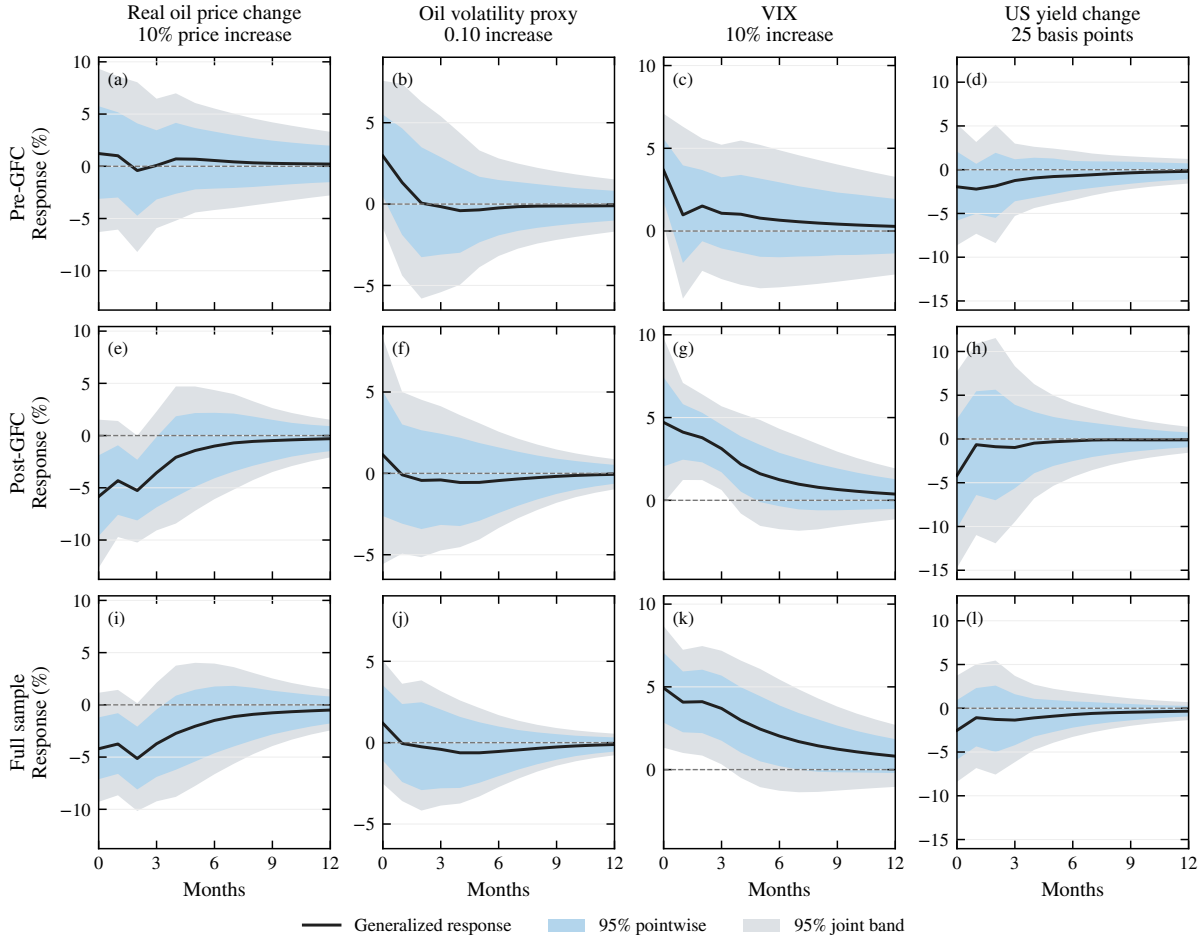
We estimate the system for March 2000–August 2008, July 2009–December 2024 and the full sample. The post-GFC dynamic window includes the pandemic, unlike the post-GFC benchmark column in Table 3. The two-lag specification balances parsimony against residual dynamics. AIC and HQ select two lags in the full and post-GFC samples, whereas BIC prefers one; all three criteria prefer one before the GFC. One-lag residual diagnostics indicate remaining autocorrelation in the full and post-GFC samples. With two lags, conventional lag-12 Portmanteau p-values are 0.494, 0.763 and 0.371 for the full, post-GFC and pre-GFC samples, respectively, and all estimated companion eigenvalues lie inside the unit circle. We use two lags in all periods for comparability and examine alternative lag orders in Appendix 5. These diagnostics support a working dynamic approximation, not proof of parameter stability or structural identification.

Generalized responses follow Pesaran and Shin (1998). For an innovation of size δ_j in variable j, the associated innovation vector is $\Sigma e_j \delta_j / \Sigma_{jj}$, where e_j selects that variable. Propagating it through equation (12) allows contemporaneously correlated news to accompany the named innovation. Responses are invariant to the ordering of variables. We use identical economic increments across periods: $\log(1.10)$ for the real oil change and log VIX, 0.10 for the annualised oil-volatility proxy, and 0.25 percentage points for the yield change. Holding these increments fixed avoids confusing regime differences with changes in innovation standard deviations. The resulting responses describe joint reduced-form comovement; they do not isolate an oil-supply shock, an oil-demand shock or an exogenous monetary-policy action.

Uncertainty is obtained from 1,999 stationary-block resamples of complete monthly regression rows, with expected block length 12 (Politis and Romano 1994). Each row retains its observed lag history. Figure 6 reports pointwise intervals and a wider band simultaneous over all four innovations and horizons 0–12 within each estimation period. Appendix 5 details the resampling, the fixed-increment normalisa-

tion and sensitivity to block length. The procedure relies on weakly stationary dependence within the estimated window; it does not eliminate the constant-parameter approximation of each VAR.

Figure 6: Joint dynamics of global conditions and average currency volatility



Notes: Rows are March 2000–August 2008, July 2009–December 2024 and the full March 2000–December 2024 sample. Lines are generalized responses from a five-variable VAR(2). Columns use the same innovation increment across rows, as indicated above each column. Responses are $100\{\exp(r_j(h)) - 1\}$, proportional changes in fitted geometric-mean FX volatility. Blue shading gives 95% pointwise intervals; grey shading gives a 95% band joint over four innovations and 13 horizons within a row, not across all three periods. Axes are shared within columns.

The oil responses differ in point estimate across periods, but do not establish a structural reversal. A positive oil innovation is associated with an immediate 1.22% increase in pre-GFC geometric-mean FX volatility and a 5.84% decrease in the post-GFC period. Neither impact interval excludes zero under joint inference. The post-GFC response is significantly negative only at horizon two, where the point estimate is −5.26%; by month six it is −1.00% and statistically imprecise. In the full sample the impact estimate is −4.20%, falling to −1.49% at month six, with zero included at every horizon. Separate period bands do not test whether pre- and post-GFC responses differ. A dedicated joint band for the post-minus-pre response difference also includes zero at every horizon for all four innovations (Appendix 5). Thus the point-estimate contrast does not establish a statistically supported cross-period difference, much less a transition from demand-driven to supply-driven oil disturbances.

The VIX response is more consistently positive. In the full sample, a 10% VIX innovation is associated with 4.93% higher geometric-mean currency volatility on impact and 2.03% at month six. The joint band excludes zero at horizons zero through three. Before the GFC only the impact response is significantly

positive; after the GFC the significant positive horizons are one through three. Oil-volatility and yield innovations have no significant responses under the joint bands in any period. These comparisons place the stronger short-horizon evidence on broad financial-risk comovement, while leaving the underlying economic shocks unidentified.

The negative oil innovation response and the imprecise conditional oil coefficient answer different questions. The generalized response permits VIX, yields and oil variability to move with an oil innovation; the baseline coefficient holds their current values fixed. Appendix Figure **??** illustrates the distinction with recursive allocations. Ordering financial variables before oil reduces the full-sample oil impact magnitude from -4.20% to -2.04% and the post-GFC magnitude from -5.84% to -2.97%. This sensitivity cautions against labelling all oil-associated comovement an autonomous oil-market effect. The recursive orderings are descriptive identification alternatives, not evidence that either set of exclusion restrictions is valid.

4.4.3 A local-projection check on propagation restrictions

A finite-order VAR restricts every horizon response through the same lag matrices. Local projections instead estimate each horizon separately (Jordà 2005). We regress future mean log volatility on the current five-variable vector and its two lags, then apply the same generalized-innovation normalisation. VAR and projection estimates use identical 284 anchor months, May 2000–December 2023; future outcomes extend through December 2024. This common sample prevents a change in observation window from masquerading as a difference between dynamic methods. The block bootstrap jointly re-estimates both procedures, including the innovation covariance, and constructs a band for their paired difference.

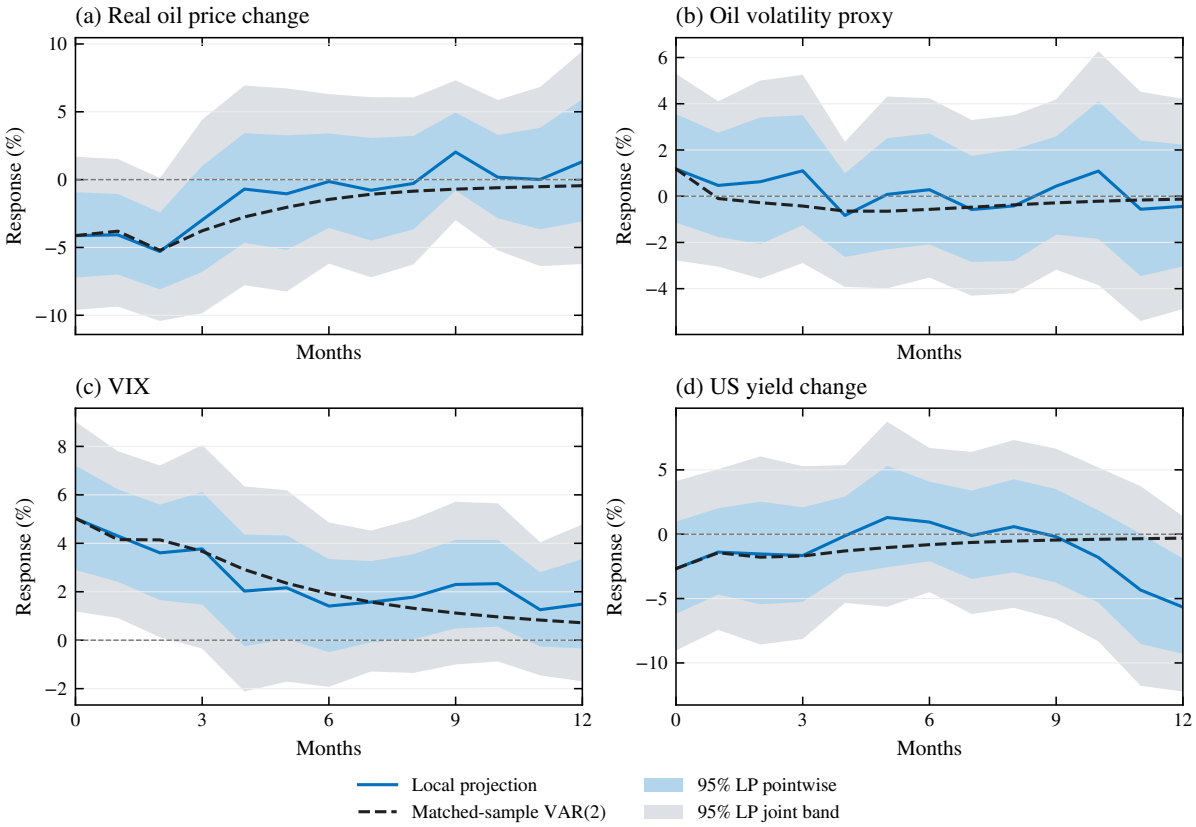
Figure 7 shows a negative oil-associated point response near impact, followed by much less precise estimates at longer horizons. No oil projection response excludes zero under joint inference. The VIX projection response is significantly positive at horizons zero through two. The joint band for the difference between the two methods includes zero at every horizon and for every innovation. This is a failure to detect disagreement, not proof that the VAR restrictions are correct: the projection intervals admit economically meaningful differences. Alternative full-sample VAR lag orders retain an insignificant oil response under joint inference, while increasing the expected block length to 24 months produces one significant negative horizon. Taken together, the dynamic checks support limited short-horizon financial-risk comovement and a qualified account of oil-associated adjustment, not a persistent causal oil effect.

5 Conclusion

Oil price movements and currency volatility share a changing global environment, but a changing estimated coefficient is not sufficient evidence of a changing economic effect. This paper studies eight advanced economies over March 2000–December 2024 using daily-based monthly FX volatility, a local-linear varying-coefficient model and a dynamic system for global conditions and mean currency volatility. The empirical design separates measurement, conditional association, cross-country heterogeneity and joint innovation responses, with uncertainty assessed over the time paths used for interpretation.

Three findings emerge. First, the common oil-change coefficient varies substantially in point estimate, yet its baseline simultaneous band includes zero throughout the interior and accommodates a constant coefficient. Alternative oil benchmarks and conditioning sets reveal some negative episodes, but none establishes a statistically supported positive-to-negative reversal. Second, the VIX association is more clearly positive and time-varying, while country oil sensitivities can depart from their cross-country average. Shared volatility episodes therefore coexist with heterogeneous conditional exposure. Third,

Figure 7: Local projections and VAR responses on an identical sample



Notes: Blue lines are local-projection responses; dashed black lines are VAR(2) responses estimated on the same 284 anchor months. Both use the fixed innovation increments in Figure 6. Shading shows 95% pointwise and joint bands for the projection responses; the joint band spans four innovations and horizons 0–12. A separate paired bootstrap band for the LP-minus-VAR difference includes zero everywhere. All responses are reduced-form and expressed as proportional changes in geometric-mean FX volatility.

the dynamic evidence places greater confidence on short-horizon financial-risk comovement than on a persistent oil response. The post-GFC oil response is negative at one horizon under joint inference; the full-sample oil response is imprecise, and local projections do not provide stronger oil-sign evidence.

These results have practical implications for currency-risk assessment. Oil-price scenarios should be evaluated together with the financial conditions accompanying them: holding VIX fixed and allowing it to move with an oil innovation represent different scenarios. Country-specific exposures also matter, since a pooled coefficient can conceal deviations from the average without explaining their economic source. Finally, risk management should retain uncertainty about both the sign and persistence of an oil-associated volatility response. A visually prominent trough in a fitted coefficient or impulse response is insufficient grounds for a stable hedging rule, and the retrospective estimates do not establish an out-of-sample forecasting improvement.

Several boundaries delimit the contribution. The panel is a fixed group of eight US-dollar exchange rates, daily sampling is an approximation to within-month risk, and the smoothing design yields its strongest inferential support away from the sample endpoints. Dynamic estimates impose a stationary constant-parameter approximation within each window, and the bootstrap conditions on bandwidth or lag choices. Most fundamentally, observed oil innovations combine supply, demand and financial news. Distinguishing those sources would require independently justified shock measures or instruments and a corresponding identification design. Within these boundaries, the evidence argues for a measured interpretation of oil–currency linkages: their conditional strength depends on the financial environment and currency composition, while universal claims about sign reversals or lasting stabilisation exceed the precision of the data.

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A. Data Construction and Sample Definitions

This appendix documents the observations underlying Sections 3 and 4.4. Its purpose is to make the volatility measures, global covariates and estimation windows reproducible. Table A1 identifies each source series. Raw observations are frozen at the 9 November 2026 retrieval date; estimation ends in December 2024. These are historical data in the retrieved vintage, not a real-time information set available to investors at each date.

Table A1: Source series and measurement conventions

Input	FRED identifier	Frequency and units
Australian dollar	`DEXUSAL`	Daily US dollars per Australian dollar
Canadian dollar	`DEXCAUS`	Daily Canadian dollars per US dollar
Japanese yen	`DEXJPUS`	Daily yen per US dollar
New Zealand dollar	`DEXUSNZ`	Daily US dollars per New Zealand dollar
Norwegian krone	`DEXNOUS`	Daily kroner per US dollar
Swedish krona	`DEXSDUS`	Daily kronor per US dollar
Swiss franc	`DEXSZUS`	Daily francs per US dollar
British pound	`DEXUSUK`	Daily US dollars per pound sterling
WTI crude oil	`POILWTIUSDM`	IMF monthly period-average US dollars per barrel
Brent crude oil	`POILBREUSDM`	IMF monthly period-average US dollars per barrel
US consumer prices	`CPIAUCSL`	BLS monthly CPI, seasonally adjusted
US ten-year yield	`GS10`	Federal Reserve monthly average, percent per annum
US industrial production	`INDPRO`	Federal Reserve monthly index, seasonally adjusted, 2017=100
VIX	`VIXCLS`	Cboe daily closing index

Notes: FX observations are Federal Reserve H.10 noon buying rates in New York. Brent and industrial production enter the sensitivity analysis. Sources: Board of Governors of the Federal Reserve System (2026a), International Monetary Fund (2026b,a), U.S. Bureau of Labor Statistics (2026), Board of Governors of the Federal Reserve System (2026c,b), Cboe (2026). Different reciprocal FX quotations leave the squared log returns and return standard deviations unchanged.

Daily FX inputs begin in December 1999. Each currency's missing quotes are removed before computing successive-observation log returns. The return is assigned to the month containing its ending quote, including when it spans a holiday. No missing quote is interpolated and no holiday is assigned a zero return. Equation (9) averages squared returns within the month and annualises at 252 trading observations. The alternative demeaned measure is $SD_{it} = \sqrt{252}\,\mathrm{sd}(r_{id} : d \in D_{it})$, using the sample standard deviation. Consequently, it differs from RV_{it} through both demeaning and the degrees-of-freedom correction.

For the monthly-return comparison, let E^m_{it} be the last available daily FX quote in month t and let $R^m_{it} = \log E^m_{it} - \log E^m_{i,t-1}$. Define

$$M_{it} = \sqrt{12}\,\mathrm{sd}(R^m_{it}, R^m_{i,t-1}, R^m_{i,t-2}). \tag{A.1}$$

The regression outcome in this exercise is $\log M_{it}$. Three complete monthly returns are available from March 2000 onward. Each standard deviation uses two degrees of freedom, and adjacent windows share two returns. The construction therefore differs from a standard deviation of daily returns in both sampling frequency and temporal overlap. The pooled arithmetic mean of M_{it} is 9.06% annualised and its pooled log correlation with RV_{it} is 0.404.

Monthly global inputs begin in 1999, ensuring that all initial differences and three-month oil-volatility windows are complete. WTI and Brent are deflated by the same US CPI; the January 2000 normalisation cancels in log changes. Brent variability uses the same three-return sample-standard-deviation formula as WTI. Industrial-production growth is $\Delta \log INDPRO_t$. VIX is aggregated by taking the arithmetic mean of available daily closes and then its logarithm; the log of an average and the average of logs are

not interchanged. Yield changes are differences in a percent-per-annum series and hence measured in percentage points.

The baseline contains 298 months for each of eight currencies. The pre-GFC window is March 2000–August 2008. The constant-coefficient post-GFC benchmark ends in January 2020 to separate it from the COVID-onward column; the dynamic post-GFC system extends from July 2009 through December 2024. September 2008–June 2009 belongs to the full sample but neither neighbouring dynamic window. The sensitivity through December 2019 contains 238 months. Local projections use anchor dates May 2000–December 2023 and future outcomes through December 2024. These distinctions determine which observations support each estimate and prevent comparisons of differently defined windows from being interpreted as formal regime tests.

B. Local Estimation, Simultaneous Inference and Graphical Diagnostics

This appendix explains how the common-regressor structure and serial dependence enter the estimates in Sections 4.1–4.3. It also provides the numerical and graphical diagnostics behind the time-path interpretation. The displays complement the main coefficient figure; repeated views of the same estimates do not constitute independent robustness tests.

B.1 Aggregation and local design

Write $D_s = (1, x_s')'$ and $w_s = K((\tau_s - \tau)/h)$ at a given target date. For any country effects satisfying $\sum_i \alpha_i = 0$, the local least-squares objective separates as

$$\begin{aligned}&\sum_{i,s} w_s\{y_{is} - \alpha_i - D_s'b_0 - D_s'b_1(\tau_s - \tau)/h\}^2 \\ &\quad = N\sum_s w_s\{\bar{y}_s - D_s'b_0 - D_s'b_1(\tau_s - \tau)/h\}^2 + \sum_{i,s} w_s\{y_{is} - \bar{y}_s - \alpha_i\}^2. \end{aligned} \tag{B.1}$$

The cross term vanishes because the within-month deviations and country effects both sum to zero. The second term contains no common coefficient. Thus the common slopes are exactly those obtained from the mean equation, even though the panel remains useful for describing country levels and heterogeneity. This equivalence does not require independent countries; their covariance affects the variance of $\bar{y}_s$.

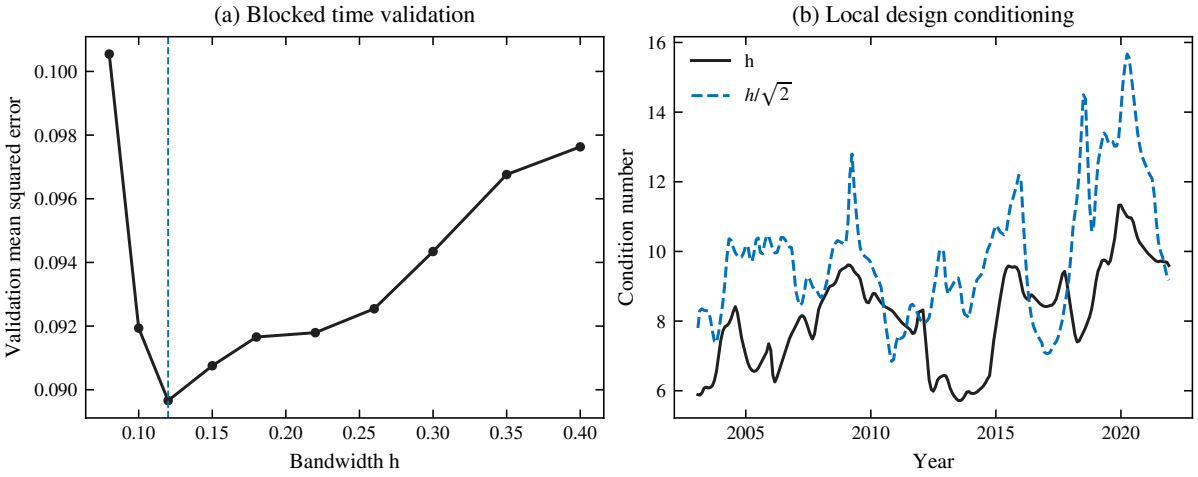
The baseline local design has ten columns: a constant, four standardised covariates, scaled local time and four covariate–time interactions. Weighted least squares is solved by singular-value decomposition, with full-rank checks at every target date. Standardisation is an affine reparameterisation and coefficients are transformed back to the economic units in equation (1). No ridge penalty enters the estimator. Figure B1 reports bandwidth validation and the conditioning of the two designs used in the bias correction.

B.2 Residual scaling and dependent multipliers

Let $H = 2H_{h/\sqrt{2}} - H_h$ map the outcome vector into the bias-corrected coefficient paths, and let S map it into fitted mean volatility, with $S_{ts} = D_t'H(t,s)$. The smoother S is generally neither symmetric nor idempotent; bias correction can also generate negative weights. Its residual variance factor under an independent homoskedastic reference model is

$$d_t = [(I - S)(I - S)']_{tt} = 1 - 2S_{tt} + \sum_s S_{ts}^2. \tag{B.2}$$

Figure B1: Bandwidth validation and local design conditioning



Notes: The validation criterion holds out consecutive six-month blocks, with a three-month exclusion margin on each side and all countries withheld together. The minimum occurs at $h = 0.12$. Condition numbers refer to the standardised weighted designs at h and $h/\sqrt{2}$ on the baseline interior. They diagnose numerical conditioning, not exogeneity.

We centre the fitted residuals and divide them by $\sqrt{d_t}$ before applying dependent multipliers. This row-norm factor is appropriate to a general linear smoother; the OLS expression $1 - S_{tt}$ is not its general variance formula. Equation (B.2) is a finite-sample scaling device based on a homoskedastic reference, not an exact variance correction under arbitrary heteroskedasticity, serial dependence or approximation bias. Temporal dependence in inference is represented by the common multipliers in equation (5).

For each draw, the same combined operator H acts on the adjusted residual–multiplier product. This retains the covariance between the two bandwidth fits in the bias correction. Multiplier standard deviations and maximum statistics are computed from these perturbations, with the maximum restricted to the specified interior. For country heterogeneity, the same residual-scaling factor and date multiplier are applied to every country; country coefficient perturbations are then centred across countries before taking the maximum over countries and dates. The resulting joint critical value is 3.716. Table B1 reports which currencies depart from the average oil slope under that criterion.

Table B1: Country heterogeneity in oil-change slopes

Country	Mean FX volatility (%)	Above mean slope	Below mean slope
Australia	11.04	0	1
Canada	7.80	0	0
Japan	9.21	5	0
New Zealand	11.75	0	0
Norway	11.47	0	0
Sweden	10.90	16	0
Switzerland	9.25	0	0
United Kingdom	8.65	4	1

Notes: Mean volatility uses all 298 months. The final two columns count interior months in which the country oil-change slope differs from the cross-country mean slope, using a single 95% maximum-t band simultaneous over all eight countries and 227 dates (critical value 3.716). All four slopes and the intercept are allowed to vary by country in this exercise. Residual multipliers are shared across countries and use the same linear-smoother residual adjustment as the aggregate analysis. Departures from the mean are not tests of a country slope against zero, nor tests of export-status mechanisms.

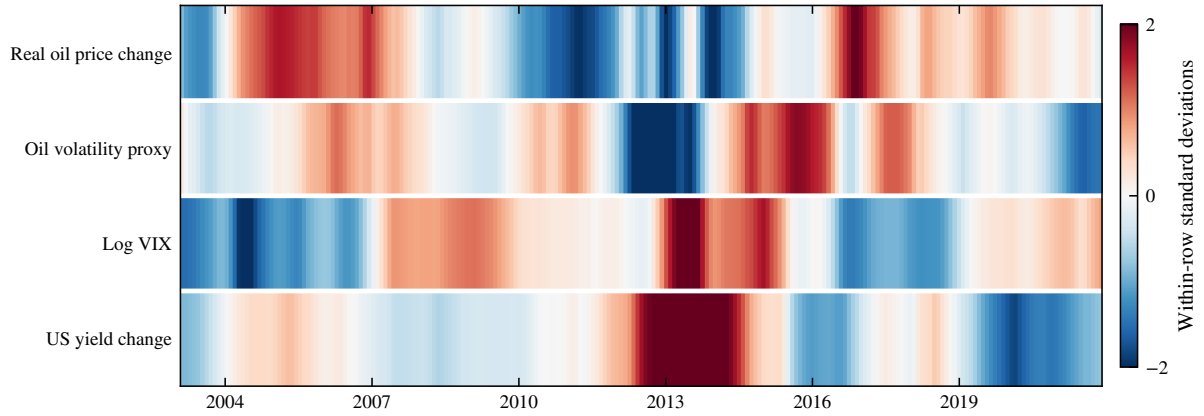
The calculations preserve the selected bandwidth and the observed design during resampling. Validity requires adequate local variation, sufficiently smooth coefficient functions and weak dependence for

the multiplier approximation. Selection uncertainty and finite-sample coverage are not eliminated by increasing the number of bootstrap draws. Computational checks confirm the aggregation identity, exact recovery of affine coefficient paths, consistent units, and nesting of pointwise intervals within simultaneous bands; these are implementation checks rather than evidence of exact coverage for the empirical data-generating process.

B.3 Reading the additional coefficient displays

Figures B2 and B3 describe the relative timing of estimated paths after centring and standardising each coefficient over the interior. A high value means that a coefficient is above its own temporal mean, not necessarily above zero. The early-2010s oil trough and the elevated VIX and yield coefficients therefore appear together without establishing a common break or a ranking of economic importance. Original units and uncertainty remain in Figure 2.

Figure B2: Relative temporal profiles of the estimated coefficients



Notes: Each coefficient is centred and divided by its own interior time-series standard deviation. The colour scale is clipped at two standard deviations for readability. Red means above that coefficient's own time average and blue means below it. Colours do not identify the sign of the raw coefficient, its economic magnitude or statistical significance. Only the interior inference region is shown.

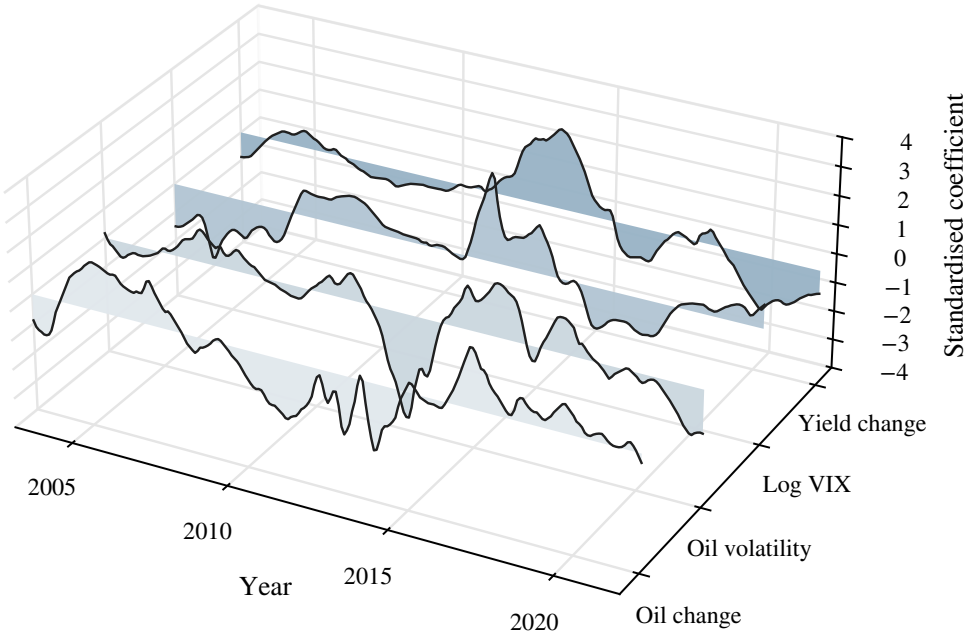
The remaining displays explain the oil-sign result. Figure B4 overlays illustrative bootstrap perturbations and contrasts month-specific significance with whole-path inference. Figure B5 places each point estimate against its pointwise lower and upper bounds. A negative interval requires the upper-bound marker to lie below the horizontal zero line; a positive interval requires the lower-bound marker to lie above it. Both displays show why a selected pointwise episode cannot substitute for a simultaneous statement about the full path.

C. Sensitivity Design and Additional Results

This appendix supports the measurement and conditioning checks in Section 4.4. Its purpose is to identify which parts of the oil result survive specific changes, while making visible the changes in estimand or observation window. Table 5 contains benchmark, measurement and control alternatives; Table C1 supplies the complete smoothing, country-omission and dependence-span results.

Except for the bandwidth experiment, the local calendar half-window is fixed at approximately 36 months. If the number of months changes from T to T', the bandwidth fraction becomes $h' = hT/T'$. The lagged-outcome model loses March 2000, while the pre-2020 sample loses the final five years. Their

Figure B3: Standardised temporal coefficient profiles



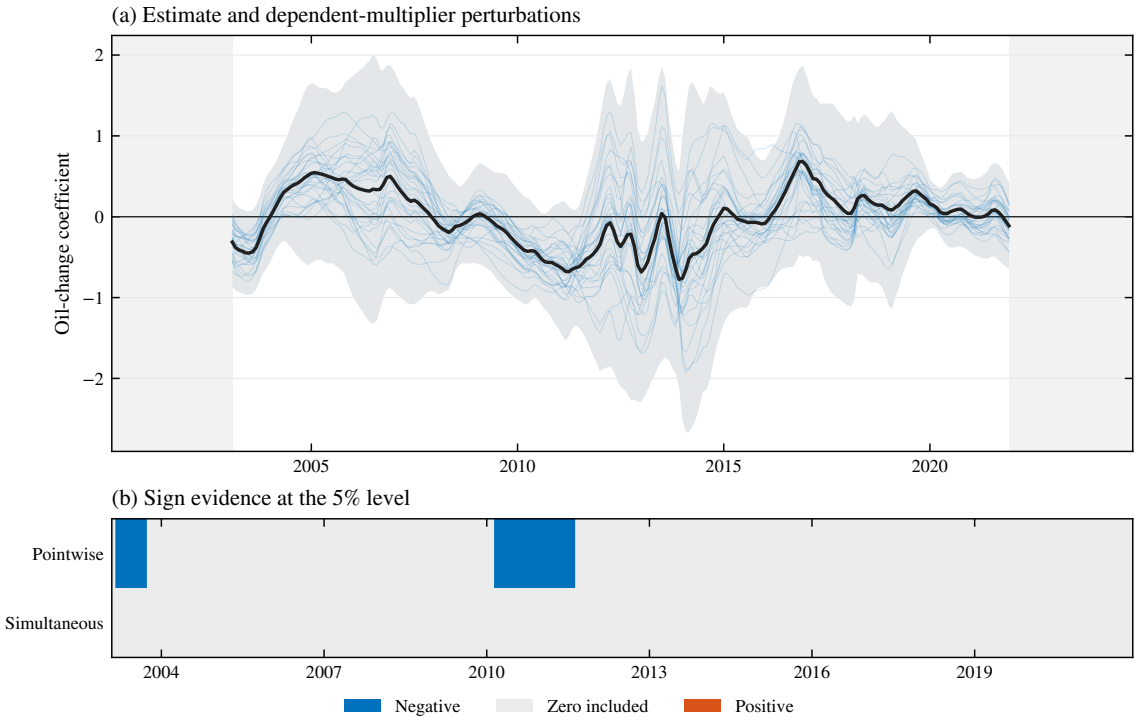
Notes: Each profile is centred and divided by its own interior standard deviation. Height measures relative temporal variation, not an economic response or a significance statistic. This three-dimensional display and Figure B2 use the same estimates.

Table C1: Smoothing, country composition and dependence-span checks

Specification	Interior months	Mean oil slope	SCB +	SCB −
Bandwidth 0.8h	241	-0.025	0	0
Bandwidth 1.2h	213	-0.018	0	0
Omit Australia	227	0.012	0	0
Omit Canada	227	-0.035	0	3
Omit Japan	227	-0.021	0	0
Omit New Zealand	227	0.040	0	0
Omit Norway	227	0.004	0	0
Omit Sweden	227	-0.044	0	0
Omit Switzerland	227	-0.017	0	0
Omit United Kingdom	227	-0.004	0	0
Dependence-span sensitivity in the baseline				
Span	Interior months	Critical value	SCB +	SCB −
6 months	227	3.249	0	0
12 months	227	3.156	0	0
18 months	227	3.280	0	0

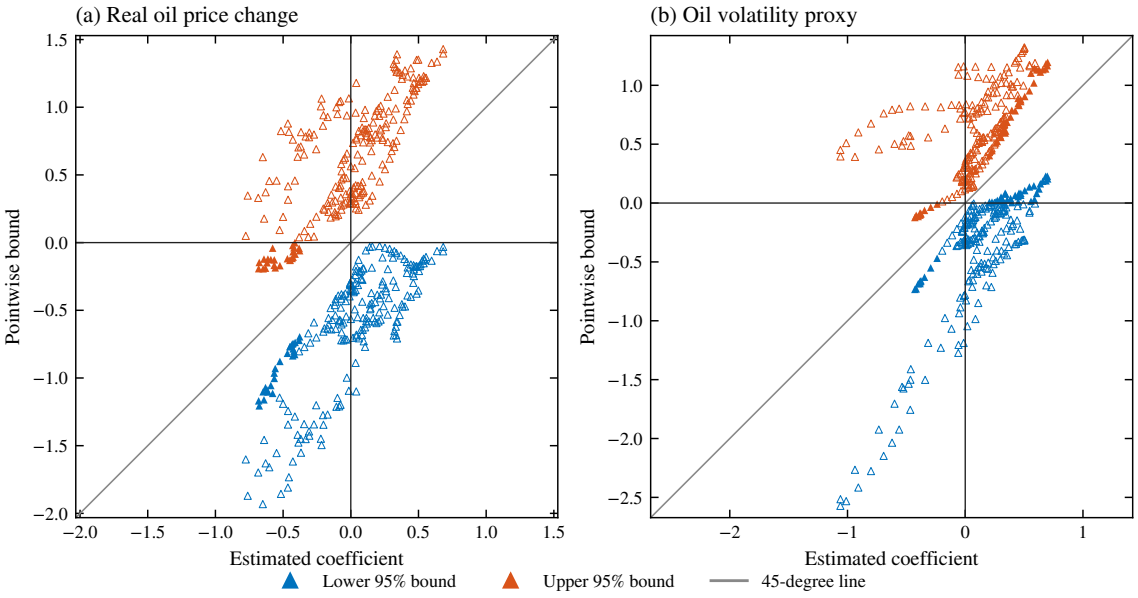
Notes: All rows have a nonempty compatible-constant set. The smaller bandwidth uses July 2002–July 2022 as its interior; the larger uses September 2003–May 2021. Country omissions retain the 227-date baseline interior and estimate the remaining seven-country average. Removing a country checks composition sensitivity, not independent time-series replication. Bands are simultaneous over time within each row; no familywise adjustment is made over the collection of specifications. The lower panel changes only the multiplier span.

Figure B4: Oil-change uncertainty and the distinction between pointwise and simultaneous evidence



Notes: Panel (a) overlays 30 illustrative centred dependent-multiplier perturbations on the estimated oil coefficient and its 95% simultaneous band. These lines are not alternative datasets, structural scenarios or forecast paths. Panel (b) records whether zero is excluded at each interior month. The pointwise row contains negative episodes; the simultaneous row does not.

Figure B5: Coefficient estimates and their pointwise interval bounds



Notes: Each pair of triangles represents one interior month: the horizontal coordinate is its estimated coefficient and the vertical coordinates are its lower and upper 95% pointwise bounds. Filled markers identify months whose pointwise interval excludes zero. The 45-degree line aids comparison only. This is a re-expression of existing interval information, not a separate statistical test.

interiors therefore differ from the baseline and are stated in the table notes. The smaller- and larger-bandwidth exercises deliberately change the window and consequently the interior. Figure 5 plots all descriptive paths on the baseline interval for visual comparison, whereas table sign counts use the appropriate formal interior for each fit.

All alternatives refit the intercept and every included slope, rather than changing only an oil coefficient in isolation. Removing VIX changes the contemporaneous information set. Adding lagged average volatility changes the short-run conditional projection. Replacing WTI with Brent changes both the price-change and oil-variability series. These contrasts can diagnose sensitivity, but they are not estimators of an unchanged structural parameter. Likewise, an omitted country changes the population average represented by the regression. The omission exercise checks whether the result is dominated by one currency; it does not create independent global shocks or test a net-oil-exporter mechanism.

No specification in Tables 5 and C1 has an empty oil constant set. This consistency concerns the absence of a rejection by a conservative band-inversion test. It does not establish equality of coefficients across specifications or prove constancy in any one specification. Negative simultaneous intervals under Brent, without VIX and after omitting Canada demonstrate the practical limit of a blanket "insignificant oil" conclusion. Their interpretation is local and conditional, with no positive significant interval supporting a sign reversal. Simultaneous coverage is calculated separately for each specification; the full set of alternative models is not covered by one omnibus critical value.

D. Dynamic Responses, Resampling and Local Projections

This appendix supplies the dynamic methods behind Section 4.4.2. It explains why the system uses one observation per month, how fixed-size innovations are propagated, how uncertainty is computed, and what the projection and ordering checks can establish. The dynamic exercise concerns average log FX volatility, while country-specific exposure is analysed separately in Section 4.3.

D.1 Lag selection, innovation scaling and horizon responses

For the VAR in equation (12), set $\Psi_0 = I$ and $\Psi_h = \sum_{\ell=1}^{\min(2,h)} A_\ell \Psi_{h-\ell}$. The generalized log-volatility response to innovation j is

$$r_j(h) = e_5' \Psi_h \frac{\Sigma e_j}{\Sigma_{jj}} \delta_j, \qquad j = 1, \ldots, 4, \quad h = 0, \ldots, 12. \tag{D.1}$$

The vector e_5 selects the mean-log-volatility equation. The ratio $\Sigma e_j / \Sigma_{jj}$ maps a specified movement in innovation j into the linearly associated contemporaneous innovation vector. Under a Gaussian innovation distribution it is the corresponding conditional expectation; more generally it provides a linear generalized-response interpretation. As in Pesaran and Shin (1998), correlated innovations are allowed and no Cholesky ordering is required for equation (D.1). This invariance is distinct from structural identification.

The increments are $\delta = (\log(1.10), 0.10, \log(1.10), 0.25)'$. All reported percentages transform log responses as $100\{\exp(r_j(h)) - 1\}$. Since $\exp(\bar{y}_t)$ is the geometric mean of the eight volatility measures, this is a proportional response in that fitted index. It is not a percentage-point movement in an exchange rate, and it need not equal a change in the conditional arithmetic expectation of volatility without additional distributional assumptions.

Table D1 presents the lag-selection and stability evidence. AIC, BIC and HQ compare orders one through six on a common within-period sample, thus the likelihood criterion is not confounded by a

changing number of observations. The main two-lag models then use all available observations after their initial two months. The full and post-GFC samples favour two lags under AIC and HQ, and the residual diagnostic improves relative to one lag. Pre-GFC evidence favours one lag; the common two-lag model uses 100 usable observations and 11 regressors per equation. This limited sample is a reason to interpret its wide bands cautiously, even though its estimated companion matrix is stable.

Table D1: Dynamic specification and bootstrap diagnostics

	Pre-GFC	Post-GFC	Full sample
Months	102	186	298
VAR(2) usable rows	100	184	296
Maximum companion eigenvalue	0.923	0.792	0.849
Portmanteau p-value, VAR(1)	0.476	0.017	< 0.001
Portmanteau p-value, VAR(2)	0.371	0.763	0.494
AIC preferred lag	1	2	2
BIC preferred lag	1	1	1
HQ preferred lag	1	2	2
Joint-band critical value	3.614	3.567	3.413
Unstable bootstrap draws	3	0	0

Notes: Pre-GFC: March 2000–August 2008; post-GFC: July 2009–December 2024; full sample: March 2000–December 2024. Each VAR contains five variables, an intercept and two lags. The conventional Portmanteau diagnostic examines residual autocorrelation through lag 12; its reference distribution is not a heteroskedasticity-robust specification test. Information criteria compare lags 1–6 on a common estimation sample within each period. The joint band controls over four innovations and horizons 0–12 within a period. All 1,999 draws are retained, including the three unstable pre-GFC estimates, because responses are evaluated only to horizon 12. Stability of the estimated VAR does not establish structural identification or parameter constancy.

D.2 Stationary-block resampling and joint bands

Let $W_t = (1, Z'_{t-1}, Z'_{t-2})'$ be the VAR regressor vector. The resampling unit is the complete regression row (W'_t, Z'_t), not a country and not an isolated residual component. A stationary bootstrap (Politis and Romano 1994) draws an initial row uniformly. At each subsequent position it starts a new uniformly chosen row with probability $1/L$, otherwise continuing to the next chronological row, with circular continuation at the endpoint. The expected run length is $L = 12$. Each selected row retains its observed lags, so concatenating blocks does not fabricate a lag relation across their boundaries. Ordinary least squares, the residual covariance and all horizon responses are recomputed in every draw.

This pairs-block implementation preserves short-range dependence in the observed regression rows, including cross-variable dependence and conditional heteroskedastic patterns under a stationary weak-dependence approximation. It does not impose a constant innovation variance by construction, but neither does it establish validity under arbitrary deterministic variance breaks or nonstationary regimes. The number of lags and the innovation increments are held fixed. The finite horizon permits computation for unstable bootstrap coefficient draws; three of the 1,999 pre-GFC draws are unstable and are retained rather than conditioning the distribution on estimated stability. All baseline full-sample and post-GFC draws are stable.

For response estimates $\widehat{r}_j(h)$ and bootstrap draws $\widehat{r}_j^{*(b)}(h)$, let $s_j(h)$ be the across-draw standard deviation. Define

$$C_{.95} = Q_{.95}\left[\max_{j=1,\ldots,4;\, h=0,\ldots,12}\left|\frac{\widehat{r}_j^{*(b)}(h) - \widehat{r}_j(h)}{s_j(h)}\right|\right]. \tag{D.2}$$

The joint log-response band is $\widehat{r}_j(h) \pm C_{.95} s_j(h)$, transformed monotonically into percentages for the

figures. Pointwise intervals use the corresponding horizon-specific quantile. Joint coverage refers to all 52 response coordinates within one period, not all periods simultaneously and not a formal cross-period difference. Table D2 reports lag-order and expected-block-length checks. The full-sample oil response remains imprecise in the baseline, one- and three-lag models, and with a six-month expected block. A 24-month block produces one negative significant horizon, illustrating that isolated significance is sensitive to dependence choices.

Table D2: Full-sample dynamic sensitivity

VAR lags	Expected block length	Oil impact (%)	Critical value	SCB +	SCB −
2	12	-4.20	3.413	0	0
1	12	-2.97	3.308	0	0
3	12	-4.39	3.406	0	0
2	6	-4.20	3.397	0	0
2	24	-4.20	3.283	0	1

Notes: Oil impact is the horizon-zero percentage response to an innovation of log(1.10) in the real oil price change. SCB columns count horizons with positive or negative responses under a 95% band joint over horizons 0–12 and all four innovations. Each row uses 1,999 stationary-block draws; all fitted full-sample models and bootstrap draws in these specifications are stable. Varying the block length changes uncertainty, not the point estimate. The 24-month block result illustrates sensitivity of isolated significance; the row comparisons do not identify a robust structural response.

D.3 Cross-period response differences

To compare the two non-overlapping windows formally, define $d_j(h) = r_j^{\text{post}}(h) - r_j^{\text{pre}}(h)$. Each replication subtracts the pre-GFC bootstrap response from an independently resampled post-GFC response, using the same fixed innovation increments in both periods. This treats cross-window sampling covariance as negligible under weak temporal dependence; excluding the crisis interval does not imply literal economic independence. Applying the joint maximum statistic to the difference yields a critical value of 3.406 and a band that contains zero at every horizon for every innovation. The comparison therefore fails to reject equality detectable with this precision. It is not an equivalence test, an endogenous break-date test or a test distinguishing oil-supply from oil-demand shocks.

D.4 Matched-sample local projections

For $h = 0, \ldots, 12$, the projection is

$$\bar{y}_{t+h} = a_h + \theta_h' Z_t + \sum_{\ell=1}^{2} \Gamma_{h\ell}' Z_{t-\ell} + \eta_{t+h,h}. \tag{D.3}$$

Here a_h is the horizon intercept, θ_h describes current-state associations, and $\Gamma_{h\ell}$ controls for two observed lag vectors. The generalized projection response is $\theta_h' \Sigma e_j \delta_j / \Sigma_{jj}$, with Σ estimated from a VAR(2) on the identical anchor dates. This is a projection counterpart to equation (D.1); it does not supply a new identifying shock. Separate horizon regressions relax the finite-order VAR propagation restrictions but generally use parameters less efficiently.

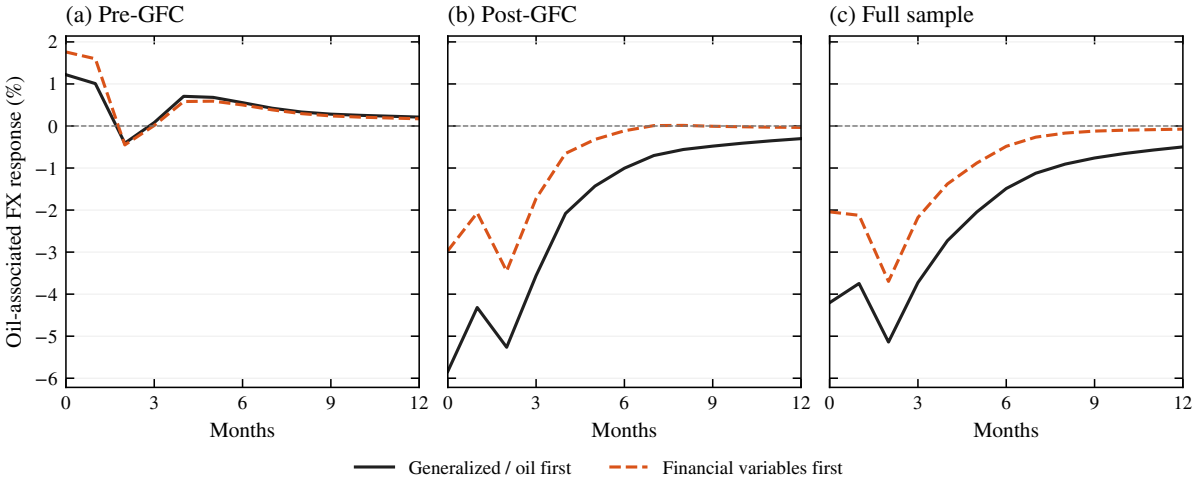
The common anchors are May 2000–December 2023, giving 284 observations at every horizon. Resampled rows jointly contain the VAR regressors and outcomes, the projection regressors and all 13 future outcomes. The same stationary row indices are used for both methods and all horizons, preserving the dependence relevant to overlapping future outcomes. Each draw re-estimates Σ and both sets of responses, so uncertainty in innovation normalisation enters the comparison. A separate maximum band applies to the paired LP-minus-VAR response difference. Its critical value is 3.546 and its band includes

zero in every coordinate. This nonrejection concerns agreement detectable with the available precision; it is not an equivalence test or validation of a causal interpretation.

D.5 Contemporaneous allocation and recursive ordering

Figure D1 compares the generalized oil response with a recursive system ordered as $(z_t, q_t, v_t, g_t, \bar{y}_t)$. In that recursive system, oil is orthogonalised after the financial variables and oil volatility. With oil ordered first, its recursive response equals the generalized response algebraically, after matching the innovation size. Their equality therefore supplies no independent corroboration. Ordering financial variables first attenuates the negative full-sample and post-GFC oil responses. This demonstrates that part of the generalized oil response travels with correlated financial innovations; it does not determine which contemporaneous causal ordering is economically correct. The figure reports point-estimate sensitivity only, without attaching structural significance to either recursive allocation.

Figure D1: Oil-response sensitivity to contemporaneous allocation



Notes: Black is the generalized oil response, identical to the oil-first recursive response at the matched innovation size. Dashed orange places log VIX, the yield change and oil variability before the real oil change, with mean log FX volatility last. Both use the same estimated VAR(2) and an oil-change increment of log(1.10). These are descriptive response allocations; no confidence bands or valid structural exclusions are asserted.